

A QUANTITATIVE FRAMEWORK FOR AD-TECH INTEGRITY

MATHEMATICAL THEORY OF FRAUD DETECTION

*Probabilistic Models, Dynamic Inference,
and Artificial Intelligence for
Lead Quality Assessment*



$$P(\text{Fraud} | E) = \frac{P(E | \text{Fraud})P(\text{Fraud})}{P(E)}$$

FOUNDATIONS · METHODS · APPLICATIONS

YURI GRANT

MATHEMATICAL THEORY OF FRAUD DETECTION

Probabilistic Models, Dynamic Inference, and Artificial Intelligence for Lead Quality Assessment

Volume I: Theory

Volume II: Systems, AI, and Applications

By Yuri Grant

Yuri Grant is a mathematician, a specialist in computer science and data analysis, and a software architect with over fifty years of experience in the fields of applied mathematics, information technology, machine learning, and decision support systems.

He began his academic career as a Professor of Mathematics and Computer Science at the State Maritime University of Ukraine, where he taught and conducted research from 1973 through the late 1990s. During this period, his work focused on applied mathematics, statistical modeling, optimization methods, and software engineering.

In the late 1990s and early 2000s, he served as a research consultant at the University of Buenos Aires, Argentina, and worked as a lead software architect for several technology companies, including Salut Uno. His work involved the design and implementation of enterprise information systems, large-scale databases, and distributed software architectures.

Since 2003, he has been a co-founder and senior technology consultant at ITSOL LLC, leading software architecture and advanced analytics projects. From 2011 to 2014, he served as Executive Director of Dentallab LLC, where he oversaw technology strategy and business development.

Since 2017, Yuriy has worked as a Research Consultant for LeadsMarket LLC, specializing in lead marketplace optimization, predictive analytics, machine learning, fraud detection, auction algorithms, and artificial intelligence. His research combines statistical modeling, operations research, optimization theory, and AI to solve complex problems in online lead generation.

He is also the founder of Complete Data Mining LLC and Robust Data Modeling LLC, companies dedicated to advanced analytics, data science consulting, and intelligent decision-support systems.

Dedication

To the engineers and data scientists who fight fraud every day—and to the honest consumers and advertisers who deserve a marketplace they can trust.

Epigraph

“Whoever is careless with the truth in small matters cannot be trusted with important matters.”

— Albert Einstein

“The marketplace is a network of trust; fraud is its corruption.”

— The Author

Table of Contents

Preface: The Problem of Fraud in Digital Marketplaces.....	20
• Why This Book Exists.....	20
• The Lead Generation Ecosystem as a Case Study.....	20
• What This Book Offers That Others Do Not.....	21
• How to Read This Book: A Reader’s Guide	21
VOLUME I — THEORY.....	21
PART I: FOUNDATIONS.....	21
Chapter 1: The Lead Generation Ecosystem.....	21
• 1.1 Historical Evolution of Lead Generation	21
• 1.2 Market Participants and Their Incentives	22
• 1.3 Business Processes and Transaction Flows.....	22
• 1.4 The Economics of Lead Trading	22
• 1.5 Ping Trees and Lead Exchanges.....	23
• 1.6 Affiliate Networks and Publisher Relationships	23
• 1.7 Sources of Uncertainty in Lead Markets.....	23
• 1.8 Sources of Fraud: A Preliminary Taxonomy	23
• 1.9 Fraud Archetypes: Running Examples.....	24
– 1.9.1 The Bot-Driven Click Fraud	24
– 1.9.2 The Traffic Laundering Ring	24
– 1.9.3 The Synthetic Identity Campaign	24
– 1.9.4 The Publisher Quality Degradation	24
• 1.10 Summary and Outlook.....	24
Chapter 2: Mathematical Representation of the Marketplace.....	24
• 2.1 Introduction: Why Formalization Matters	24
• 2.2 Entities: The Building Blocks.....	25
– 2.2.1 Leads	25
– 2.2.2 Publishers.....	26
– 2.2.3 Consumers.....	26
– 2.2.4 Devices.....	27

–	2.2.5 Sessions.....	28
–	2.2.6 Buyers.....	28
–	2.2.7 Campaigns.....	29
–	2.2.8 Auctions	29
–	2.2.9 Conversions	30
•	2.3 The Marketplace as a Relation Graph.....	30
•	2.4 Probabilistic Events and Temporal Evolution	31
•	2.5 The Master State-Space Model.....	32
–	2.5.1 Hidden States $Z(i,t)$	32
–	2.5.2 Observations $O(i,t)$	33
–	2.5.3 Fraud Indicators $F(i,t)$	33
–	2.5.4 State Transitions and Emissions	34
•	2.6 Complete Notation and Symbol Table.....	34
•	2.7 The Master Formalism in Words	36
•	2.8 Summary and Outlook.....	36
•	2.9 Bibliographic Notes	37

Chapter 3: Probability Theory for Lead Systems.....38

•	3.1 Random Variables and Sample Spaces	38
•	3.2 Conditional Probability and Independence	40
•	3.3 Bayes' Theorem and Its Implications	41
•	3.4 Joint Distributions and Factorizations.....	43
•	3.5 Hidden Variables and Latent States.....	46
•	3.6 Expectation, Variance, and Moments	48
•	3.7 Information and Entropy	50
•	3.8 Common Distributions in Fraud Modeling.....	52
•	3.9 Worked Applications in Lead Marketplaces.....	56
•	3.10 Summary and Outlook.....	59
•	3.11 Bibliographic Notes	60

PART II: THE MATHEMATICAL THEORY OF FRAUD AND TRUST..... 61

Chapter 4: Fraud as a Latent Stochastic Process.....61

•	4.1 Formal Definition of Fraud.....	61
•	4.2 Fraud Probability, Severity, and Confidence	64
•	4.3 Legitimate Behavior vs. Fraudulent Behavior	66
•	4.4 Fraud as a Stochastic Process.....	67
•	4.5 Temporal Dependence in Fraud Processes.....	69
•	4.6 The Fraud Signal-to-Noise Problem.....	71
•	4.7 Examples of Fraud Processes in Lead Systems	73
•	4.8 Summary and Outlook.....	76
•	4.9 Bibliographic Notes	77

Chapter 5: Taxonomy of Fraud	78
• 5.1 Human Fraud	78
• 5.2 Bot Fraud	80
• 5.3 Affiliate Fraud	84
• 5.4 Identity Fraud	86
• 5.5 Synthetic Identities	87
• 5.6 Conversion Fraud	89
• 5.7 Publisher Fraud	90
• 5.8 Traffic Laundering	92
• 5.9 Fraud Rings	94
• 5.10 Mixed and Hybrid Fraud	96
• 5.11 Fraud Classes as a Graph	96
• 5.12 Summary and Outlook	97
• 5.13 Bibliographic Notes	98
Chapter 6: Observable Evidence Space	99
• 6.1 Introduction: What Can We See?	99
• 6.2 IP Address and Geolocation	101
• 6.3 Device Fingerprints and Browser Signals	103
• 6.4 Mouse Dynamics and Input Behavior	105
• 6.5 Timing and Session Patterns	107
• 6.6 Typing Dynamics	109
• 6.7 Email and Phone Signals	110
• 6.8 Historical Data and Relationships	112
• 6.9 Conversion Outcomes and Post-Sale Data	113
• 6.10 Formal Representation of the Observation Vector $O(i,t)$	114
• 6.11 Feature Engineering and Preprocessing	115
• 6.12 Data Quality and Missing Values	116
• 6.13 Summary and Outlook	117
• 6.14 Bibliographic Notes	118
Chapter 7: The Ground-Truth Problem	119
• 7.1 Why Ground Truth Is So Difficult in Fraud	119
• 7.2 Chargebacks, Disputes, and Buyer Rejections	120
– 7.2.1 Noise in Labels	121
– 7.2.2 Delay in Labels	122
– 7.2.3 Bias in Labels	123
• 7.3 Selection Bias in Confirmed-Fraud Data	124
• 7.4 Positive-Unlabeled Learning	125
– 7.4.1 The PU Learning Framework	125
– 7.4.2 Two-Step Approaches	126
– 7.4.3 Direct Learning Approaches	127
• 7.5 Weakly Supervised Learning	128

– 7.5.1 Inexact Supervision	128
– 7.5.2 Inaccurate Supervision	129
– 7.5.3 Incomplete Supervision	130
• 7.6 Label Validation and Consensus.....	131
• 7.7 Consequences for Supervised Methods.....	132
• 7.8 Summary and Outlook.....	134
• 7.9 Bibliographic Notes	135

Chapter 8: Hidden Quality and Trust States 135

• 8.1 Introduction: The Unobservable Fundamentals	136
• 8.2 Publisher Quality States.....	137
• 8.3 Campaign Quality States.....	139
• 8.4 Consumer Quality States	140
• 8.5 Buyer Quality States.....	141
• 8.6 Device and Session Quality States	142
• 8.7 Joint Estimation Across Entity Types	143
• 8.8 Quality States as a Graph-Coupled Markov Process	145
• 8.9 Operational Definitions of Quality States	146
• 8.10 Transition Diagrams Between States	147
• 8.11 Instantiating the Master Model.....	149
• 8.12 Summary and Outlook.....	150
• 8.13 Bibliographic Notes	150

PART III: SEQUENTIAL AND BAYESIAN STATE ESTIMATION 151

Chapter 9: Bayesian Fraud Inference..... 151

• 9.1 Introduction: The Posterior Distribution	151
• 9.2 Prior Specification.....	152
– 9.2.1 Informative Priors.....	153
– 9.2.2 Non-Informative Priors.....	153
– 9.2.3 Hierarchical Priors.....	154
• 9.3 Likelihood Functions for Fraud Models	155
• 9.4 Sequential Updating.....	156
• 9.5 Hierarchical Bayesian Models.....	157
– 9.5.1 Two-Level Hierarchies	157
– 9.5.2 Multi-Level Hierarchies	158
• 9.6 Approximate Inference Methods	159
– 9.6.1 Markov Chain Monte Carlo	159
– 9.6.2 Variational Inference	160
– 9.6.3 Particle Filters	160
– 9.6.4 Expectation Propagation	161
• 9.7 Posterior Predictive Checks	161
• 9.8 Uncertainty Quantification	162

• 9.9 Summary and Outlook.....	163
• 9.10 Bibliographic Notes	164
Chapter 10: Dynamic Bayesian Networks and Hidden Markov Models	165
• 10.1 Introduction: State-Space Models	165
• 10.2 Dynamic Bayesian Networks	166
– 10.2.1 General Formulation	166
– 10.2.2 Graph Structure and Conditional Independence.....	166
– 10.2.3 Inference in DBNs.....	167
• 10.3 Hidden Markov Models	168
– 10.3.1 The Single-Chain Special Case.....	168
– 10.3.2 Forward-Backward Algorithm	168
– 10.3.3 Viterbi Algorithm	169
– 10.3.4 Baum-Welch Algorithm	170
• 10.4 Publisher Quality as Hidden State.....	171
• 10.5 Observations: Conversion, Acceptance, Revenue, Complaints	172
• 10.6 Dependencies Among Publisher, Campaign, Device, Consumer, Buyer	173
• 10.7 Learning DBN Parameters	174
• 10.8 Model Selection and Structure Learning	175
• 10.9 Complete DBN Diagram and Worked Example.....	176
• 10.10 Summary and Outlook	177
• 10.11 Bibliographic Notes	178
Chapter 11: Sequential and Change-Point Detection.....	179
• 11.1 Introduction: When Fraud Regimes Change.....	179
• 11.2 Cumulative Sum (CUSUM) Control Charts	180
– 11.2.1 Basic CUSUM.....	180
– 11.2.2 Multivariate CUSUM.....	181
• 11.3 Sequential Probability Ratio Test (SPRT)	182
– 11.3.1 Wald's SPRT	182
– 11.3.2 Composite Hypotheses	183
– 11.3.3 Optimal Stopping	183
• 11.4 Bayesian Online Change-Point Detection.....	184
– 11.4.1 Changepoint Models	184
– 11.4.2 Online Inference	184
– 11.4.3 Run-Length Probability	185
• 11.5 Detecting Sudden Attacks.....	186
• 11.6 Detecting Gradual Degradation.....	187
• 11.7 Multi-Stream Change Detection	188
• 11.8 Information Criteria for Model Selection	189
• 11.9 Comparison of Methods	190
• 11.10 Summary and Outlook	191
• 11.11 Bibliographic Notes	191

PART IV: STATISTICAL AND MACHINE LEARNING MODELS 192

Note: This Part presents the emission model $P_{\phi}(O|Z)$ and transition model P_{θ} as parametric and non-parametric families, as introduced in Chapter 2. 192

Chapter 12: Parametric Emission Models 192

• 12.1 Introduction: $P_{\phi}(O Z)$ Under Known Distributions	192
• 12.2 Linear Regression Models	193
• 12.3 Logistic Regression	194
– 12.3.1 Binary Classification	194
– 12.3.2 Multi-Class Classification	195
– 12.3.3 Regularization	195
• 12.4 Generalized Linear Models	196
– 12.4.1 Exponential Family	196
– 12.4.2 Link Functions	196
– 12.4.3 Model Selection	197
• 12.5 Survival Models	197
– 12.5.1 Time-to-Event Modeling	197
– 12.5.2 Cox Proportional Hazards	198
– 12.5.3 Accelerated Failure Time	198
• 12.6 Mixed-Effects Models	199
– 12.6.1 Fixed and Random Effects	199
– 12.6.2 Hierarchical Modeling	199
• 12.7 Model Assumptions and Diagnostics	200
• 12.8 Parameter Estimation: MLE and MAP	201
• 12.9 Summary and Outlook	201
• 12.10 Bibliographic Notes	202

Chapter 13: Non-Parametric Emission Models 203

• 13.1 Introduction: Distribution-Free Approaches	203
• 13.2 Decision Trees	204
– 13.2.1 Classification and Regression Trees	204
– 13.2.2 Splitting Criteria	205
– 13.2.3 Pruning	206
• 13.3 Random Forest	206
– 13.3.1 Bagging and Ensemble Methods	206
– 13.3.2 Feature Importance	207
– 13.3.3 Out-of-Bag Error	207
• 13.4 Gradient Boosting	208
– 13.4.1 XGBoost	208
– 13.4.2 LightGBM	209
– 13.4.3 CatBoost	209
• 13.5 Feature Engineering for Fraud Detection	210
– 13.5.1 Transformations and Interactions	210

–	13.5.2 Domain-Specific Features.....	211
–	13.5.3 Aggregation Features	212
•	13.6 Interpretability of Tree-Based Models.....	214
–	13.6.1 Feature Importance	214
–	13.6.2 Partial Dependence Plots.....	214
–	13.6.3 Tree Surrogates	215
•	13.7 Hyperparameter Tuning.....	215
•	13.8 Summary and Outlook.....	216
•	13.9 Bibliographic Notes	217
Chapter 14: Sequence Emission Models		218
•	14.1 Introduction: Temporal Dependencies in $O(i,t)$	218
•	14.2 Feed-Forward Neural Networks.....	219
–	14.2.1 Architecture Design	219
–	14.2.2 Activation Functions.....	220
–	14.2.3 Training and Optimization	220
•	14.3 Recurrent Neural Networks	221
–	14.3.1 Vanilla RNNs	221
–	14.3.2 Long Short-Term Memory (LSTM)	222
–	14.3.3 Gated Recurrent Units (GRU).....	223
–	14.3.4 Bidirectional RNNs.....	223
•	14.4 Transformer Models.....	224
–	14.4.1 Attention Mechanisms	224
–	14.4.2 Self-Attention and Multi-Head Attention.....	224
–	14.4.3 Positional Encoding	225
–	14.4.4 Application to Sequential Fraud Data.....	225
•	14.5 Temporal Embeddings	226
•	14.6 Contrastive Learning.....	227
–	14.6.1 Siamese Networks	227
–	14.6.2 Supervised Contrastive Learning	227
–	14.6.3 Applications to Fraud Detection	227
•	14.7 Sequence Modeling for Lead Fraud	228
–	14.7.1 Session Sequences.....	228
–	14.7.2 User Journeys	228
–	14.7.3 Publisher Trajectories.....	229
•	14.8 Summary and Outlook.....	229
•	14.9 Bibliographic Notes	230
Chapter 15: Representation Learning for Fraud		230
•	15.1 Introduction: Learning Useful Representations	230
•	15.2 Autoencoders	232
–	15.2.1 Vanilla Autoencoders	232
–	15.2.2 Variational Autoencoders.....	232
–	15.2.3 Denoising Autoencoders.....	233
•	15.3 Embeddings for Categorical Variables	234

– 15.3.1 Entity Embeddings.....	234
– 15.3.2 Learned Embeddings	234
• 15.4 Self-Supervised Learning.....	235
• 15.5 Transfer Learning for Fraud.....	237
• 15.6 Summary and Outlook.....	239
• 15.7 Bibliographic Notes	239

PART V: NETWORK AND INFORMATION-THEORETIC METHODS..... 240

Chapter 16: Graph Theory and Graph Neural Networks..... 240

• 16.1 Introduction: The Marketplace as a Graph.....	240
• 16.2 Graph Representation.....	242
– 16.2.1 Nodes: Consumer, Publisher, Buyer, Device, IP, Phone	242
– 16.2.2 Edges: Interactions and Relationships	242
– 16.2.3 Edge Types and Weights.....	243
– 16.2.4 Temporal Graphs	244
• 16.3 Graph Properties.....	245
– 16.3.1 Connected Components	245
– 16.3.2 Centrality Measures.....	245
– 16.3.3 Clustering Coefficient.....	246
– 16.3.4 Graph Distance.....	247
• 16.4 Communities, Cycles, and Fraud Rings	247
– 16.4.1 Community Detection	247
– 16.4.2 Cycle Detection	248
– 16.4.3 Anomaly Detection in Graphs	249
• 16.5 Graph Neural Networks	250
– 16.5.1 Graph Convolutional Networks (GCN)	250
– 16.5.2 Graph Attention Networks (GAT)	251
– 16.5.3 GraphSAGE	251
– 16.5.4 Graph Isomorphism Networks (GIN).....	252
– 16.5.5 Temporal GNNs	252
• 16.6 Graph Representation Learning	253
– 16.6.1 Node Embeddings.....	253
– 16.6.2 Edge Embeddings	253
– 16.6.3 Graph Embeddings.....	254
– 16.6.4 Node2Vec and DeepWalk	254
• 16.7 Applications to Fraud Detection	255
– 16.7.1 Suspicious Community Detection.....	255
– 16.7.2 Link Prediction for Collusion	256
– 16.7.3 Node Classification for Fraud.....	256
• 16.8 Summary and Outlook.....	257
• 16.9 Bibliographic Notes	258

Chapter 17: Information Theory for Fraud Diagnostics	259
• 17.1 Introduction: Information as a Diagnostic Tool	259
• 17.2 Entropy and Its Properties.....	260
• 17.3 Conditional Entropy	261
• 17.4 Mutual Information.....	262
– 17.4.1 Definition and Properties	262
– 17.4.2 Estimation Methods.....	262
– 17.4.3 Feature Selection	263
• 17.5 Kullback-Leibler Divergence.....	264
– 17.5.1 Definition and Properties	264
– 17.5.2 Asymmetry and Symmetrization.....	264
– 17.5.3 Applications	265
• 17.6 Jensen-Shannon Divergence.....	265
• 17.7 Transfer Entropy	266
– 17.7.1 Directed Information Flow.....	266
– 17.7.2 Causal Inference	266
• 17.8 Active Information	267
• 17.9 Applications to Fraud Diagnostics	268
– 17.9.1 Traffic Diversity.....	268
– 17.9.2 Publisher Entropy.....	269
– 17.9.3 Behavioral Entropy	269
– 17.9.4 Change Detection in Distributions	270
• 17.10 Summary and Outlook	270
• 17.11 Bibliographic Notes	271

PART VI: BEHAVIORAL MATHEMATICS271

Note: This Part presents concrete instantiations of $P_{\phi}(O|Z)$ for behavioral signals.....271

Chapter 18: Human Behavioral Models.....	271
• 18.1 Introduction: The Signature of Human Behavior.....	271
• 18.2 Mouse Dynamics	272
– 18.2.1 Movement Trajectories	272
– 18.2.2 Velocity and Acceleration.....	274
– 18.2.3 Curvature and Smoothness.....	275
• 18.3 Typing Dynamics	275
– 18.3.1 Keystroke Timing.....	275
– 18.3.2 Keystroke Pressure	276
– 18.3.3 Error Patterns	277
• 18.4 Scrolling and Navigation	278
• 18.5 Touchscreen Gestures.....	279
– 18.5.1 Swipes and Taps.....	279
– 18.5.2 Multi-Touch Patterns	280
• 18.6 Human Variability and Authenticity	281

• 18.7 Statistical Models of Human Behavior	282
• 18.8 Summary and Outlook	283
• 18.9 Bibliographic Notes	283

Chapter 19: Bot Modeling..... 284

• 19.1 Introduction: The Adversary's Automation	284
• 19.2 Bot Behavior Characteristics.....	285
• 19.3 Finite-State Automata for Bots.....	286
• 19.4 Generative Bot Models	288
• 19.5 Adversarial Bots.....	289
– 19.5.1 Evasive Bots.....	289
– 19.5.2 Adaptive Bots	290
– 19.5.3 Distributed Bots	291
• 19.6 Synthetic Traffic Generation	292
• 19.7 Bot Evolution and Co-Evolution	293
– 19.7.1 Bot Arms Races.....	293
– 19.7.2 Detecting Bot Evolution	294
• 19.8 Summary and Outlook.....	295
• 19.9 Bibliographic Notes	295

Chapter 20: Behavioral Biometrics and Distance Metrics..... 296

• 20.1 Introduction: Measuring Behavioral Similarity.....	296
• 20.2 Distance Metrics for Behavioral Signals	297
– 20.2.1 Euclidean Distance.....	297
– 20.2.2 Mahalanobis Distance.....	298
– 20.2.3 Manhattan Distance	298
• 20.3 Sequence Alignment.....	299
– 20.3.1 Needleman-Wunsch Algorithm.....	299
– 20.3.2 Smith-Waterman Algorithm.....	300
• 20.4 Dynamic Time Warping	301
– 20.4.1 Basic DTW	301
– 20.4.2 Fast DTW.....	302
– 20.4.3 Derivative DTW	302
• 20.5 Neural Embeddings for Behavior	303
– 20.5.1 Behavioral Embeddings	303
– 20.5.2 Metric Learning.....	304
• 20.6 Behavioral Classification	305
• 20.7 Anomaly Detection in Behavior	307
• 20.8 Summary and Outlook.....	308
• 20.9 Bibliographic Notes	308

PART VII: DECISION, GAME, AND MECHANISM DESIGN THEORY..... 309

Chapter 21: Decision Theory and Cost-Sensitive Classification 309

• 21.1 Introduction: Decisions Under Uncertainty.....	309
---	-----

• 21.2 Expected Loss and Risk	310
• 21.3 Cost-Sensitive Classification.....	312
– 21.3.1 Cost Matrices	312
– 21.3.2 Cost-Sensitive Learning.....	312
– 21.3.3 MetaCost.....	313
• 21.4 Optimal Rejection Thresholds	313
• 21.5 Utility Functions.....	315
– 21.5.1 Expected Utility	315
– 21.5.2 Utility Elicitation	315
• 21.6 Sequential Decision Making.....	316
• 21.7 Decision-Theoretic Fraud Filtering	317
• 21.8 Summary and Outlook	318
• 21.9 Bibliographic Notes	319
Chapter 22: Auction Theory and Mechanism Design for Ping Trees	319
• 22.1 Introduction: Ping Trees as Auction Mechanisms	319
• 22.2 Auction Theory Fundamentals	321
– 22.2.1 Types of Auctions.....	321
– 22.2.2 Revenue Equivalence	321
– 22.2.3 Bidder Valuation Models.....	322
• 22.3 Cascading Ping-Tree Auctions	323
– 22.3.1 Structure and Dynamics.....	323
– 22.3.2 Sequential Auction Models.....	324
– 22.3.3 Information Asymmetry	325
• 22.4 Revenue-Optimal Auctions Under Fraud Risk	326
– 22.4.1 Myerson’s Revenue-Optimal Auction	326
– 22.4.2 Extension to Fraud-Affected Auctions.....	326
– 22.4.3 Optimal Reserve Prices.....	327
• 22.5 Incentive Compatibility in Fraud-Affected Markets.....	327
– 22.5.1 Truthful Bidding.....	327
– 22.5.2 Fraudulent Manipulation.....	328
• 22.6 Fraud-Robust Mechanism Design	328
– 22.6.1 Robust Optimization.....	328
– 22.6.2 Bayesian Robust Mechanisms.....	329
• 22.7 Practical Implementation in Lead Markets	330
• 22.8 Summary and Outlook	331
• 22.9 Bibliographic Notes	331
Chapter 23: Reinforcement Learning for Adaptive Filtering and Pricing	332
• 23.1 Introduction: Adaptation in Fraudulent Environments	332
• 23.2 Markov Decision Processes	333
– 23.2.1 State, Action, Reward	333
– 23.2.2 Transition and Reward Functions.....	334
– 23.2.3 Discounting	334
• 23.3 Value Functions and Bellman Equations	335

• 23.4 Model-Based Reinforcement Learning.....	336
– 23.4.1 Planning.....	336
– 23.4.2 Dyna Architecture	336
• 23.5 Model-Free Reinforcement Learning.....	337
– 23.5.1 Q-Learning.....	337
– 23.5.2 SARSA.....	337
– 23.5.3 Deep Q-Networks (DQN)	338
• 23.6 Policy Gradient Methods.....	339
– 23.6.1 REINFORCE	339
– 23.6.2 Actor-Critic Methods.....	339
– 23.6.3 Proximal Policy Optimization (PPO).....	340
• 23.7 Adaptive Filtering of Fraud.....	340
• 23.8 Adaptive Pricing in Fraud-Affected Markets	341
• 23.9 Adaptive Publisher Trust.....	342
• 23.10 Exploration vs. Exploitation.....	342
– 23.10.1 Epsilon-Greedy.....	342
– 23.10.2 Upper Confidence Bound (UCB).....	343
– 23.10.3 Thompson Sampling.....	343
• 23.11 Off-Policy Evaluation	343
• 23.12 Summary and Outlook.....	344
• 23.13 Bibliographic Notes	345
Chapter 24: Multi-Agent Fraud Games	346
• 24.1 Introduction: Strategic Interactions in the Marketplace.....	346
• 24.2 Game Theory Fundamentals	347
– 24.2.1 Normal-Form Games.....	347
– 24.2.2 Extensive-Form Games.....	347
– 24.2.3 Bayesian Games.....	348
• 24.3 Nash Equilibrium.....	349
– 24.3.1 Pure Strategy Equilibrium.....	349
– 24.3.2 Mixed Strategy Equilibrium.....	349
– 24.3.3 Computational Approaches	350
• 24.4 Players in the Fraud Game	351
– 24.4.1 Fraudsters	351
– 24.4.2 Publishers	351
– 24.4.3 Buyers	352
– 24.4.4 The Marketplace Platform.....	352
• 24.5 Strategic Fraud Behavior	353
– 24.5.1 Rational Fraud	353
– 24.5.2 Collusion and Fraud Rings	353
• 24.6 Mechanism Design in Fraudulent Environments.....	354
– 24.6.1 The Revelation Principle.....	354
– 24.6.2 Implementation Theory	354
• 24.7 Repeated Games and Reputation	355

–	24.7.1 Folk Theorems	355
–	24.7.2 Reputation Mechanisms	356
•	24.8 Evolutionary Game Theory	356
•	24.9 Summary and Outlook	357
•	24.10 Bibliographic Notes	358
Chapter 25: Adversarial Robustness of Detection Models.....		359
•	25.1 Introduction: The Cat-and-Mouse Game	359
•	25.2 Adversarial Machine Learning.....	360
–	25.2.1 Threat Models	360
–	25.2.2 Attack Surfaces	361
•	25.3 Evasion Attacks on Fraud Classifiers	361
–	25.3.1 Gradient-Based Attacks	361
–	25.3.2 Optimization-Based Attacks	362
–	25.3.3 Decision-Based Attacks	363
•	25.4 Adversarial Examples	363
–	25.4.1 Fast Gradient Sign Method (FGSM).....	363
–	25.4.2 Projected Gradient Descent (PGD).....	364
–	25.4.3 Carlini-Wagner Attack	364
•	25.5 Robust Statistics Against Strategic Manipulation	365
–	25.5.1 Huber Loss.....	365
–	25.5.2 Trimmed Estimators.....	365
–	25.5.3 Median-of-Means	366
•	25.6 Adversarial Training.....	366
–	25.6.1 Data Augmentation	366
–	25.6.2 Robust Optimization.....	367
–	25.6.3 Defensive Distillation	367
•	25.7 Certified Robustness	367
–	25.7.1 Randomized Smoothing	367
–	25.7.2 Certification via Lipschitz Bounds.....	368
•	25.8 Detection of Adversarial Examples.....	368
•	25.9 Domain-Specific Defenses for Lead Systems	369
•	25.10 Summary and Outlook	370
•	25.11 Bibliographic Notes	370

PART VIII: EVALUATION AND CAUSAL INFERENCE.....371

Chapter 26: Metrics Under Class Imbalance and Label Noise.....		371
•	26.1 Introduction: The Challenge of Evaluation	371
•	26.2 Confusion Matrix and Derived Metrics.....	372
–	26.2.1 Accuracy and Its Limitations.....	372
–	26.2.2 Precision and Recall.....	373
–	26.2.3 F1 and F-Beta Scores	374
–	26.2.4 Specificity and Negative Predictive Value.....	374

• 26.3 Imbalanced Classification.....	375
– 26.3.1 The Class Imbalance Problem.....	375
– 26.3.2 Sampling Techniques.....	375
– 26.3.3 Cost-Sensitive Approaches.....	376
• 26.4 PR-AUC vs. ROC-AUC.....	377
– 26.4.1 ROC Analysis.....	377
– 26.4.2 Precision-Recall Analysis.....	377
– 26.4.3 Comparison and Recommendations.....	378
• 26.5 Calibration of Posterior Fraud Probabilities.....	378
– 26.5.1 Calibration Plots (Reliability Diagrams).....	378
– 26.5.2 Expected Calibration Error (ECE).....	379
– 26.5.3 Maximum Calibration Error (MCE).....	379
– 26.5.4 Brier Score Decomposition.....	380
• 26.6 Metrics Under Label Noise.....	380
• 26.7 A/B Testing Under Adversarial Conditions.....	381
– 26.7.1 Design of Experiments.....	381
– 26.7.2 Non-Stationary Environments.....	381
– 26.7.3 Sequential Testing.....	382
• 26.8 Summary and Outlook.....	382
• 26.9 Bibliographic Notes.....	383
Chapter 27: Causal Inference for Measuring Fraud’s Economic Impact.....	383
• 27.1 Introduction: Causality vs. Correlation.....	383
• 27.2 The Causal Inference Framework.....	385
– 27.2.1 Potential Outcomes.....	385
– 27.2.2 Average Treatment Effect.....	385
– 27.2.3 Conditional Average Treatment Effect.....	386
• 27.3 Uplift Modeling.....	386
– 27.3.1 Treatment Effects in Fraud Detection.....	386
– 27.3.2 Meta-Learners: S-Learner, T-Learner, X-Learner.....	386
– 27.3.3 Causal Forests.....	388
• 27.4 Synthetic Control Methods.....	388
– 27.4.1 The Synthetic Control Approach.....	388
– 27.4.2 Placebo Tests.....	389
– 27.4.3 Permutation Inference.....	389
• 27.5 Difference-in-Differences for Campaign-Level Interventions.....	390
– 27.5.1 Basic Setup.....	390
– 27.5.2 Parallel Trends Assumption.....	390
– 27.5.3 Event Studies.....	391
• 27.6 Instrumental Variables for Unobserved Confounding.....	391
• 27.7 Pearl’s Do-Calculus.....	392
– 27.7.1 Causal Graphs.....	392
– 27.7.2 Identifying Causal Effects.....	392
• 27.8 Causal Discovery.....	393

– 27.8.1 Learning Causal Structure	393
– 27.8.2 Constraint-Based Methods.....	393
– 27.8.3 Score-Based Methods.....	394
• 27.9 Distinguishing Correlation-Based Signals from Causal Drivers	394
• 27.10 Applications to Fraud Economics	395
– 27.10.1 Measuring Fraud’s Revenue Impact	395
– 27.10.2 Evaluating Anti-Fraud Interventions	395
• 27.11 Summary and Outlook	396
• 27.12 Bibliographic Notes	396

PART IX: SYNTHESIS397

Chapter 28: Mathematical Theory of Trust 397

• 28.1 Introduction: Trust as a Mathematical Object.....	397
• 28.2 Trust as a Stochastic Process	398
– 28.2.1 Continuous Estimation	398
– 28.2.2 Prediction.....	399
– 28.2.3 Confidence Intervals.....	399
• 28.3 Trust as a Potential Function	400
• 28.4 Identifiability of Hidden-State Estimators.....	401
– 28.4.1 When Is Z Identifiable?	401
– 28.4.2 Parameter Identifiability	401
• 28.5 Consistency and Convergence	402
– 28.5.1 Asymptotic Consistency	402
– 28.5.2 Convergence Rates	402
– 28.5.3 Posterior Concentration.....	403
• 28.6 Spectral Properties of Trust	403
• 28.7 The Information-Theoretic View of Trust.....	404
• 28.8 Trust Propagation in Networks.....	405
• 28.9 Summary and Outlook.....	406
• 28.10 Bibliographic Notes	407

Chapter 29: Unified Theory of Fraud 407

• 29.1 Introduction: Integration Across Disciplines.....	407
• 29.2 Probability Theory: The Foundation.....	409
• 29.3 Graph Theory: The Structure	410
• 29.4 Information Theory: The Signal.....	411
• 29.5 Optimization: The Computation	412
• 29.6 Machine Learning: The Estimation.....	413
• 29.7 Artificial Intelligence: The Agency	414
• 29.8 Game Theory: The Interaction.....	415
• 29.9 A Unified Algorithm	416
– 29.9.1 The Unified Fraud Detection Framework	416
– 29.9.2 Implementation Considerations	416
• 29.10 Theorems and Guarantees.....	417

• 29.11 Summary and Outlook	418
• 29.12 Bibliographic Notes	419
Chapter 30: Open Research Problems	420
• 30.1 Introduction: The Frontiers of Fraud Research.....	420
• 30.2 Publisher State Estimation Under Adversarial Manipulation.....	421
• 30.3 Fraud Diffusion and Contagion	422
– 30.3.1 Spread of Fraud Through Networks.....	422
– 30.3.2 Epidemic Models of Fraud.....	423
• 30.4 Self-Healing Marketplaces.....	424
– 30.4.1 Automated Recovery Mechanisms.....	424
– 30.4.2 Resilient Market Design	425
• 30.5 Autonomous Marketplaces	425
– 30.5.1 Fully Autonomous Fraud Detection	425
– 30.5.2 AI Market Operators.....	426
• 30.6 Quantum-Resistant Trust Models	427
– 30.6.1 Post-Quantum Cryptography	427
– 30.6.2 Quantum-Safe Authentication	428
• 30.7 Fairness and Bias in Fraud Detection.....	429
• 30.8 Interpretability and Explainability.....	429
• 30.9 Multi-Modal Fraud Detection.....	430
• 30.10 Summary and Outlook.....	431
• 30.11 Bibliographic Notes	432
APPENDICES.....	434
Appendix A: Notation and Symbol Glossary	434
• A.1 General Notation	434
• A.2 Entities and Relations	435
• A.3 State Space Notation.....	436
• A.4 Observation Space Notation	436
• A.5 Probability Notation	437
• A.6 Graph Theory Notation.....	438
• A.7 Information Theory Notation.....	439
• A.8 Optimization Notation	439
• A.9 Machine Learning Notation	440
Appendix B: Probability Theory	440
• B.1 Basic Probability	440
• B.2 Random Variables.....	441
• B.3 Probability Distributions	441
• B.4 Expectation and Moments.....	442
• B.5 Conditional Probability.....	443

• B.6 Bayes' Theorem	443
• B.7 Law of Total Probability	444
• B.8 Inequalities	444
Appendix C: Linear Algebra	444
• C.1 Vectors and Matrices	444
• C.2 Matrix Operations	445
• C.3 Eigenvalues and Eigenvectors.....	446
• C.4 Matrix Decompositions.....	446
• C.5 Norms and Inner Products	446
Appendix D: Graph Theory.....	447
• D.1 Basic Definitions.....	447
• D.2 Graph Types	447
• D.3 Graph Properties.....	448
• D.4 Graph Algorithms.....	448
• D.5 Spectral Graph Theory.....	449
Appendix E: Information Theory	449
• E.1 Entropy.....	449
• E.2 Mutual Information	450
• E.3 Divergence Measures.....	450
• E.4 Information-Theoretic Inequalities	451
Appendix F: Optimization Methods.....	451
• F.1 Convex Optimization.....	451
• F.2 Gradient Descent	451
• F.3 Stochastic Optimization	452
• F.4 Constrained Optimization	452
• F.5 Bayesian Optimization	453
Appendix G: Python Primer	453
• G.1 Basic Python.....	453
• G.2 NumPy.....	454
• G.3 Pandas	454
• G.4 Scikit-Learn	455
• G.5 PyTorch.....	456
• G.6 TensorFlow.....	457
Appendix H: SQL Examples.....	458
• H.1 Basic Queries	458
• H.2 Aggregate Functions.....	458

• H.3 Window Functions	459
• H.4 Joins	459
• H.5 Subqueries	460
Appendix I: Statistical Tables.....	460
• I.1 Normal Distribution	460
• I.2 t-Distribution	461
• I.3 Chi-Square Distribution	461
• I.4 F-Distribution	462
Appendix J: Dataset Descriptions	462
• J.1 Public Fraud Datasets	462
• J.2 Lead Generation Datasets	462
• J.3 Synthetic Data Generation	463
• J.4 Data Format and Schema	463
Appendix K: Proofs of Key Theorems	464
• K.1 Bayesian Optimality	464
• K.2 Consistency of Estimators	464
• K.3 Identifiability Results	464
• K.4 Convergence Rates	465
• K.5 Asymptotic Normality	465

Supplementary Materials

Online Companion Website: RobustDataModeling.com

Author's Note on Completeness

This two-volume monograph represents the first comprehensive mathematical treatment of fraud detection in lead generation systems. The unified formalism introduced in Chapter 2 provides a consistent foundation for the diverse methods presented across 36 chapters and 11 Parts.

The structure is designed to support multiple reading paths:

- Theoretical researchers may focus on Parts I–V and IX
- Applied researchers may focus on Parts IV–VII
- Industry practitioners may focus on Parts X–XI
- Graduate students may work through the entire text sequentially

Each chapter includes:

- Clear learning objectives
- Motivating examples
- Formal mathematical development
- Worked examples
- Bibliographic notes

The appendices provide the necessary background for readers from diverse disciplines, making the book accessible to statisticians, computer scientists, economists, and engineers alike.

Preface

The Problem of Fraud in Digital Marketplaces

Digital marketplaces have transformed commerce, yet they face a persistent and growing threat: fraud. In the lead generation industry alone, fraud costs billions of dollars annually, undermines trust, and distorts markets. This book offers a comprehensive mathematical treatment of fraud detection in lead generation systems—a problem that serves as a microcosm for fraud in all digital marketplaces.

Why This Book Exists

Existing approaches to fraud detection are fragmented across disciplines and often ad-hoc. Statisticians focus on inference; computer scientists on algorithms; economists on incentives; and practitioners on heuristics. This book unifies these perspectives into a single mathematical framework. The organizing thesis is straightforward but powerful: a lead generation platform is a dynamic probabilistic system in which every participant carries a hidden quality state that evolves over time and must be inferred from noisy observations. Fraud is a function of that state. This book provides the first comprehensive mathematical treatment of this problem, integrating probability theory, graph theory, information theory, optimization, machine learning, artificial intelligence, and game theory into a coherent framework.

The Lead Generation Ecosystem as a Case Study

Lead generation is an ideal case study because it exhibits all the key features of digital marketplaces: multiple agents with conflicting incentives, hidden information, strategic behavior, dynamic evolution, and complex network structure. Understanding fraud in lead generation provides insights applicable to online advertising, e-commerce, insurance, banking, and beyond.

What This Book Offers That Others Do Not

This book offers:

- **A Unified Mathematical Formalism**—The master state-space model (introduced in Chapter 2) provides a common language for all methods.
- **Interdisciplinary Integration**—Seven disciplines integrated into a single framework.
- **Theory-Practice Bridge**—Volume I develops the theory; Volume II implements it.
- **Comprehensive Coverage**—36 chapters, 11 parts, and 11 appendices.
- **Rigorous Treatment**—Theorems, proofs, and guarantees where appropriate.

How to Read This Book

Volume I develops the mathematical theory (Parts I-IX). It is intended for researchers and statisticians who want to understand the foundations. Volume II covers artificial intelligence methods and computational implementation (Parts X-XI). It is intended for practitioners building production systems. The appendices provide background material for readers from diverse disciplines. The book is designed to be accessible to graduate students and professionals with a background in mathematics, statistics, or computer science.

Chapter 1: The Lead Generation Ecosystem

This chapter serves as the book's foundation, introducing the complex economic and operational realities of the lead generation industry. It is designed to orient the reader—whether a statistician or a practitioner—to the domain where the mathematical theory of fraud detection will be applied. It establishes that a lead is not mere contact information, but rather a package of intent, permission, and timing with significant commercial value .

1.1 Historical Evolution of Lead Generation

This section traces the journey of the industry from its origins to its current data-driven form.

- **Pre-Internet Era:** Lead generation existed in the form of direct mail, telemarketing, and trade show “leads” (business cards).
- **The Dot-Com Boom:** The advent of the internet created a more measurable environment. Banner ads and search engines like Google allowed for the first forms of digital inquiry capture.
- **The Rise of Affiliate Marketing:** Platforms like Commission Junction (now CJ Affiliate) and later Google Ads allowed publishers to monetize traffic by driving conversions, formalizing the “pay-for-performance” model.
- **Specialization and the Ping Tree:** As the ecosystem grew, dedicated lead generation companies emerged, and the ping tree was developed as a real-time auction mechanism to sell leads instantly to the highest bidding buyer. This became a cornerstone of modern lead trading.

1.2 Market Participants and Their Incentives

The “lead economy” operates as a multi-tier marketplace . The incentives of these participants can align, creating value, or conflict, creating opportunities for fraud.

- **Tier 1: Lead Generators (Publishers/Affiliates):** These entities own media or buy traffic (e.g., via Google Ads, Facebook) to drive consumers to landing pages. Their primary incentive is to maximize the **volume** of leads generated while minimizing the **cost per lead (CPL)** . They profit from the spread between the CPL paid to the traffic source and the price at which they sell the lead.
- **Tier 2: Aggregators and Distributors (Brokers/Networks):** These sit between publishers and buyers. **Brokers** purchase leads from many sources, aggregate them, and resell them to a network of buyers, taking inventory and credit risk. **Networks and Exchanges** connect publishers and buyers for real-time transactions. Their incentive is to efficiently route a lead to the buyer who values it most, taking a fee. Brokers face margin compression from both sides .
- **Tier 3: Lead Buyers:** These are the end-customers who convert leads into business, such as insurance carriers, mortgage lenders, and solar installers. Their incentive is to acquire high-quality leads at a cost low enough to generate a positive return on investment (ROI) . The value of a lead decays rapidly over time .
- **Supporting Actors:** This group includes **Technology Providers** (CRM, distribution platforms like boberdoo or LeadsPedia), **Data Providers** (verification and consent services like TrustedForm), and **Ad Networks**.

1.3 Business Processes and Transaction Flows

This section provides a step-by-step walkthrough of the journey of a lead through the ecosystem .

- **Consumer Action:** A consumer researching a product (e.g., auto insurance) fills out a form on a publisher’s landing page. The form captures contact info and qualifying attributes (e.g., credit score, property value) .
- **Consent Capture & Validation:** The publisher must obtain explicit consumer consent to be contacted and document this process to comply with regulations like the TCPA .
- **Lead Transmission:** The publisher sends the lead data to a distributor or directly to a buyer. This is often done through a platform’s API in milliseconds.
- **Auction/Routing:** The lead is routed through a ping tree.
- **Sale & Purchase:** A buyer purchases the lead and the lead is assigned to a sales agent.
- **Conversion:** The sales agent contacts the consumer to close a sale.
- **Settlement & Feedback:** The buyer pays for the lead (less returns or allowances), and provides conversion data back to the distributor/publisher.

1.4 The Economics of Lead Trading

The financial mechanics are critical to understanding the pressures that drive fraud.

- **The Float Rule:** Time between paying publishers and collecting from buyers is cash flow sensitive. The “60-day float rule” is a well-known survival heuristic in lead generation .

- **Margins: Gross Margins** for a broker might be 25-40%, but after operational realities like returns (often 8-15%), float costs, and bad debt, **Net Margins** rarely exceed 15-18% .
- **Quality-Volume Paradox:** Tactics that increase lead volume often degrade quality. This creates a fundamental tension .
- **Trust Degradation:** Buyer relationships have a lifecycle, often degrading after 3-6 months if quality slips .
- **The Pareto Reality:** 20% of sources generate 80% of quality leads, and a failure to manage concentration can be disastrous .

1.5 Ping Trees and Lead Exchanges

A **ping tree** is a specific type of real-time lead auction used to route leads to buyers .

- **Operation:** When a lead is generated, it is “pinged” (a preliminary request) to a network of buyers, often sequentially. Buyers respond with a bid indicating how much they would pay for the lead. The lead is then “posted” to the winning buyer. The “ping/post” architecture is central to lead distribution .
- **Incentives:** Its efficiency allows publishers to access many buyers and buyers to filter leads. However, this opacity creates a significant fraud risk and is a key focus of the mathematical theory in Volume I.

1.6 Affiliate Networks and Publisher Relationships

This section expands on the relationship between Tiers 1 and 2.

- **Affiliate Networks:** Platforms that connect independent publishers (affiliates) with advertisers, providing tracking, payment, and fraud monitoring.
- **Publisher Risk:** Affiliates and publishers operate with significant platform dependency (e.g., Google policy changes), which can instantly kill a business .
- **Distributor Risk:** Distributors require sophisticated infrastructure to manage relationships on both sides and ensure quality .

1.7 Sources of Uncertainty in Lead Markets

Uncertainty is an inherent feature of the system.

- **Quality Uncertainty:** A buyer cannot perfectly know the quality of a lead until after purchase.
- **Behavioral Uncertainty:** The actions of consumers, publishers, and buyers are unpredictable.
- **Adversarial Uncertainty:** The presence of sophisticated fraudsters creates a strategic and dynamic adversarial environment.

1.8 Sources of Fraud: A Preliminary Taxonomy

This section provides an initial, high-level categorization.

- **Identity and Data Fraud:** Synthetic identities, fake personal information.
- **Traffic Source Fraud:** Bot-driven clicks, traffic laundering .
- **Quality Fraud:** Misrepresenting lead quality to get higher prices.

1.9 Fraud Archetypes: Running Examples

These concrete examples provide a backdrop for the theory developed later in the book.

- **1.9.1 The Bot-Driven Click Fraud:** A publisher uses a botnet to generate fraudulent clicks on ads or form submissions, artificially inflating lead volume. Approximately 37% of web traffic is attributed to bots .
- **1.9.2 The Traffic Laundering Ring:** A sophisticated ring uses multiple websites, shell companies, and ad networks to hide the origin of fraudulent traffic.
- **1.9.3 The Synthetic Identity Campaign:** Fraudsters generate completely fake consumer profiles (synthetic identities) to submit “leads.”
- **1.9.4 The Publisher Quality Degradation:** A legitimate publisher finds that its lead quality is degrading over time, perhaps due to a change in its traffic sources. This could be unintentional, a result of “quality-volume paradox” .

1.10 Summary and Outlook

The chapter concludes with a high-level summary of the interconnected, uncertain, and economically driven nature of the lead generation ecosystem.

A Note to the Reader: This chapter establishes the “ground truth” of the lead generation industry. The sophisticated mathematical methods described in subsequent chapters are built on this foundation of incentives, uncertainty, and fraud risk. Understanding that a lead is “intent captured at the moment of expression, packaged with permission to make contact, and sold within a time window where that intent still has commercial value” is essential to appreciate the value and the vulnerability of the system .

Chapter 2: Mathematical Representation of the Marketplace

2.1 Introduction: Why Formalization Matters

This section establishes the philosophical and practical necessity of a rigorous mathematical framework for fraud detection.

The Problem with Informal Reasoning

- Fraud detection in industry is often approached through heuristics, rules of thumb, and ad-hoc models
- These approaches fail to capture the complex dependencies, temporal dynamics, and strategic interactions in the ecosystem
- Informal reasoning leads to:
 - Inconsistent detection across different fraud types
 - Inability to reason systematically about detection performance
 - Difficulty in identifying theoretical limits of detection

- Poor generalization to new fraud patterns

The Benefits of Formalization

- **Precision:** Every concept has an unambiguous definition
- **Composability:** Complex models can be built from simpler components
- **Verifiability:** Claims can be proven or disproven mathematically
- **Transferability:** Results apply across different implementations
- **Identifiability:** We can determine what can and cannot be learned from data
- **Optimality:** We can derive theoretically optimal detectors

What This Chapter Accomplishes

- Introduces a unified notation used throughout both volumes
- Defines the fundamental objects of the lead marketplace
- Establishes the master state-space model that underpins all subsequent chapters
- Creates a common language for statisticians, computer scientists, and practitioners

The Central Insight

A lead generation marketplace is fundamentally a **dynamic probabilistic system** in which every participant carries a **hidden quality state** that evolves over time and must be inferred from **noisy observations**. Fraud is a **function of this hidden state**.

2.2 Entities: The Building Blocks

This section defines the atomic units of the marketplace—the entities that carry states, generate observations, and interact with one another.

2.2.1 Leads

Definition: A lead is the fundamental unit of exchange—a package of consumer information, intent signals, and permission to contact.

Mathematical Representation:

- A lead is not a static entity; it is a **temporally indexed object** with a lifecycle
- Let L_i denote the i -th lead in the system
- Each lead is characterized by:
 - **Creation time:** $t_i^{(creation)}$
 - **Attributes:** A vector of features (demographics, product interest, etc.)
 - **Source:** The publisher or channel that generated it
 - **Buyer:** The entity that ultimately purchases it
 - **Outcome:** Whether it converts into a sale
 - **Quality:** A hidden state that determines its true value

Lifecycle of a Lead:

- **Genesis:** A consumer expresses interest (form submission, click, call)
- **Enrichment:** Additional data is appended (IP, device, behavioral signals)
- **Routing:** The lead traverses the ping tree, receiving bids
- **Sale:** The lead is purchased by a buyer
- **Contact:** The buyer's sales team reaches out
- **Conversion:** The consumer becomes a customer (or not)
- **Feedback:** Conversion outcome is reported back to the ecosystem

Why Leads Matter Mathematically:

- Leads are the primary source of observations $O(i, t)$
- Lead quality is the central hidden state we aim to infer
- Lead lifecycles create the temporal structure for state estimation

2.2.2 Publishers

Definition: Publishers are entities that generate leads by driving traffic to landing pages or forms. They are the primary suppliers in the lead marketplace.

Mathematical Representation:

- Let P_j denote the j -th publisher in the system
- Each publisher is characterized by:
 - **Identity:** Unique identifier, domain, business name
 - **Traffic sources:** Where they acquire visitors (organic, paid, social, etc.)
 - **Lead volume:** Rate of lead generation
 - **Quality:** Hidden state representing the trustworthiness and value of their leads
 - **Relationship history:** Interactions with buyers and distributors

Publisher Dynamics:

- Publishers enter and exit the marketplace
- Publishers change their traffic sources over time
- Publisher quality evolves (improves, degrades, or fluctuates)
- Publishers may be legitimate, fraudulent, or somewhere in between

Why Publishers Matter Mathematically:

- Publisher quality is a primary hidden state $Z_P(t)$
- Publishers' strategic behavior is modeled via game theory
- Publisher networks form graphs for community detection

2.2.3 Consumers

Definition: Consumers are the end-users who express interest and become leads. They are the ultimate source of value in the ecosystem.

Mathematical Representation:

- Let C_k denote the k -th consumer in the system
- Each consumer is characterized by:
 - **Identity:** Person identifier (often pseudonymous)
 - **Demographics:** Age, location, income, etc.
 - **Behavior:** Browsing patterns, form completion, click-throughs
 - **History:** Past interactions with publishers and buyers
 - **Fraud indicator:** Whether the consumer is real or synthetic

Consumer Dynamics:

- Consumers may interact with multiple publishers
- Consumer behavior varies naturally and systematically
- Synthetic identities have behavioral patterns distinct from real consumers

Why Consumers Matter Mathematically:

- Consumer behavior is a source of observations $O(i, t)$
- Consumer-fraud indicators (synthetic identities) are a key detection target
- Consumer-publisher-buyer relationships form a graph

2.2.4 Devices

Definition: Devices are the physical or virtual endpoints through which consumers interact with publishers.

Mathematical Representation:

- Let D_l denote the l -th device in the system
- Each device is characterized by:
 - **Fingerprint:** A composite identifier (IP, user-agent, screen resolution, fonts, etc.)
 - **Type:** Desktop, mobile, tablet, or bot (headless browser)
 - **Behavior:** Timing patterns, request headers, JavaScript execution
 - **History:** Past sessions and associations

Device Dynamics:

- Devices may be shared among multiple consumers
- Devices may be virtualized or emulated by bots
- Device fingerprints may be spoofed or rotated

Why Devices Matter Mathematically:

- Device fingerprints are a key observation source
- Device graphs reveal fraud rings (shared devices)
- Bot detection relies on device behavior patterns

2.2.5 Sessions

Definition: A session is a contiguous sequence of interactions between a consumer (or bot) and a publisher.

Mathematical Representation:

- Let S_m denote the m-th session in the system
- Each session is characterized by:
 - **Start time:** $t_m^{(start)}$
 - **End time:** $t_m^{(end)}$
 - **Duration:** $\Delta t = t^{(end)} - t^{(start)}$
 - **Events:** Sequence of clicks, page views, form fills
 - **Outcome:** Whether the session generated a lead
 - **Device:** D_l (if identifiable)

Session Dynamics:

- Sessions may be legitimate human behavior or bot-generated
- Session patterns reveal intent and authenticity
- Sessions connect consumers, devices, and publishers

Why Sessions Matter Mathematically:

- Sessions are the observation window for behavioral models
- Session sequences are input to sequence models (LSTMs, Transformers)
- Session anomalies are strong fraud indicators

2.2.6 Buyers

Definition: Buyers are entities that purchase leads from publishers or distributors.

Mathematical Representation:

- Let B_n denote the n-th buyer in the system
- Each buyer is characterized by:
 - **Identity:** Unique identifier
 - **Vertical:** Industry (insurance, mortgage, solar, etc.)
 - **Bidding behavior:** How they price leads in auctions
 - **Quality requirements:** Minimum acceptable lead quality
 - **Conversion rate:** How effectively they convert leads to sales
 - **Fraud sensitivity:** How they respond to suspected fraud

Buyer Dynamics:

- Buyers enter and exit the market
- Buyer bids change based on quality observations
- Buyers may be defrauded by low-quality leads

Why Buyers Matter Mathematically:

- Buyer feedback is a source of delayed, noisy labels
- Buyer behavior is modeled in auction theory
- Buyer-publisher relationships are edges in the graph

2.2.7 Campaigns

Definition: A campaign is a structured set of marketing activities with specific parameters, targets, and budgets.

Mathematical Representation:

- Let Cmp_o denote the o -th campaign in the system
- Each campaign is characterized by:
 - **Targeting:** Geographic, demographic, behavioral
 - **Budget:** Total spend and daily caps
 - **Creative:** Ad copy, landing page design
 - **Pricing:** Cost-per-lead (CPL) or cost-per-click (CPC)
 - **Performance:** Historical conversion rates and ROI

Campaign Dynamics:

- Campaigns are launched, paused, and terminated
- Campaign performance changes over time
- Fraudsters may target specific campaigns

Why Campaigns Matter Mathematically:

- Campaigns are the context for lead generation
- Campaign performance is a source of observations
- Campaigns influence the conditional distribution of lead quality

2.2.8 Auctions

Definition: An auction is a mechanism for determining which buyer purchases a lead.

Mathematical Representation:

- Let A_q denote the q -th auction in the system
- Each auction is characterized by:
 - **Type:** Sequential (ping tree) or simultaneous
 - **Participants:** Set of buyers invited to bid
 - **Bids:** Monetary offers from buyers
 - **Winner:** The buyer with the winning bid
 - **Price:** The amount paid for the lead

Auction Dynamics:

- Auctions are instantaneous events

- Bids reflect buyer expectations of lead quality
- Auction design affects fraud incentives

Why Auctions Matter Mathematically:

- Auction theory (Chapter 22) is central to understanding ping trees
- Auctions determine prices, which affect fraud profitability
- Auction design can mitigate fraud

2.2.9 Conversions

Definition: A conversion occurs when a lead becomes a paying customer.

Mathematical Representation:

- Let $Conv_r$ denote the r-th conversion in the system
- Each conversion is characterized by:
 - **Time:** When the conversion occurred
 - **Amount:** Revenue generated
 - **Lead:** Which lead converted
 - **Buyer:** Who sold to the consumer

Conversion Dynamics:

- Conversions occur with a delay after lead purchase
- Conversion rates vary by lead source, campaign, and buyer
- Conversions are the ultimate signal of lead quality

Why Conversions Matter Mathematically:

- Conversions are the “ground truth” for lead quality
- Conversion data provides delayed, sparse labels
- Conversion prediction is a key modeling task

2.3 The Marketplace as a Relation Graph

This section defines the mathematical structure of the marketplace as a graph.

The Graph Representation:

$$G = (E, R)$$

Where:

- $E = \{e_1, e_2, \dots, e_N\}$ is the set of all entities
- $R \subseteq E \times E$ is the set of relations between entities

Entity Types: Each entity e_i has a type $\tau(e_i) \in \{\text{Lead, Publisher, Consumer, Device, Session, Buyer, Campaign, Auction, Conversion}\}$

Relation Types:

- **Generates:** Publisher → Lead (a publisher generated a lead)
- **Belongs to:** Lead → Consumer (a lead belongs to a consumer)
- **Originates from:** Lead → Device (a lead originated from a device)
- **Occurs in:** Lead → Session (a lead occurred in a session)
- **Purchased by:** Lead → Buyer (a lead was purchased by a buyer)
- **Part of:** Lead → Campaign (a lead was generated as part of a campaign)
- **Pinged in:** Lead → Auction (a lead was auctioned)
- **Results in:** Lead → Conversion (a lead resulted in a conversion)
- **Operated by:** Publisher → Affiliate Network (a publisher operates through a network)
- **Submits bid in:** Buyer → Auction (a buyer submitted a bid in an auction)

Graph Properties:

- The graph is **dynamic**—entities and relations evolve over time
- The graph is **heterogeneous**—multiple entity and relation types
- The graph is **sparse**—not all entities are connected
- The graph is **temporally ordered**—edges have timestamps

Why the Graph Matters:

- Fraud rings appear as dense subgraphs (communities)
- Quality propagates through the graph
- Graph algorithms (Chapter 16) leverage the graph structure

2.4 Probabilistic Events and Temporal Evolution

This section introduces the probabilistic framework for modeling the marketplace's evolution.

The Marketplace as a Stochastic Process: The entire marketplace evolves as a stochastic process indexed by time t . The state of the marketplace at time t is the collection of all entity states.

Discrete vs. Continuous Time:

- For theoretical simplicity, we use discrete time steps $t = 1, 2, 3, \dots$
- In practice, this corresponds to events such as:
 - Each lead generation event
 - Each auction event
 - Each conversion event
- The time step is the minimum resolution at which we observe events

Probabilistic Events: Each event in the marketplace is a probabilistic outcome:

- **Lead generation:** Consumer C_k generates lead L_i with probability p_{lead}
- **Bid submission:** Buyer B_n bids $\$b$ for lead L_i with probability p_{bid}
- **Auction outcome:** Buyer B_n wins auction with probability proportional to their bid
- **Conversion:** Lead L_i converts to sale with probability p_{conv}

Temporal Dependencies: Events are not independent over time:

- A lead's conversion probability depends on when it was generated
- Publisher quality evolves over time
- Fraudsters adapt their behavior based on detection

The Temporal Model:

$$P(\text{Marketplace State at } t + 1 \mid \text{History up to } t)$$

This conditional distribution is the foundation for:

- Bayesian filtering (Chapter 9)
 - Sequential detection (Chapter 11)
 - Reinforcement learning (Chapter 23)
-

2.5 The Master State-Space Model

This is the **central contribution** of Chapter 2—the unified formalism that anchors the entire book.

The Formal Structure:

$$\begin{aligned} Z_{i,t+1} &\sim P_{\theta}(\cdot \mid Z_{i,t}, \{Z_{j,t}\}_{j \in \mathcal{N}(i)}) && \text{(State Transition)} \\ O_{i,t} &\sim P_{\phi}(\cdot \mid Z_{i,t}) && \text{(Emission/Observation)} \\ F_{i,t} &= g(Z_{i,t}) && \text{(Fraud as a Function of State)} \end{aligned}$$

The unifying task of the book is:

$$P(Z_{1:n,t} \mid O_{1:n,1:t}) \quad \text{(Posterior Inference)}$$

2.5.1 Hidden States $Z(i, t)$

Definition: $Z(i, t)$ is the hidden state of entity i at time t .

State Space:

- $Z(i, t) \in \mathcal{S}$
- $\mathcal{S}$ can be:
 - **Discrete:** $\mathcal{S} = \{\text{Excellent, Normal, Degrading, Suspicious, Fraudulent}\}$
 - **Continuous:** $\mathcal{S} = \mathbb{R}^d$ (a vector of quality scores)
 - **Hybrid:** Mixture of discrete and continuous

Interpretation for Different Entities:

- **Publisher:** Quality of traffic, trustworthiness
- **Consumer:** Likelihood of legitimate behavior, conversion probability
- **Device:** Legitimacy of the device (human vs. bot)
- **Campaign:** Quality of the campaign's targeting and creative
- **Lead:** The lead's true quality (pre-purchase evaluation)

Why States Are Hidden:

- We cannot directly observe quality
- We must infer it from noisy observations
- The hidden-state formulation is the foundation of Bayesian inference

2.5.2 Observations $O(i, t)$

Definition: $O(i, t)$ is the observation vector associated with entity i at time t .

Observation Space:

- $O(i, t) \in \mathcal{O}$
- $\mathcal{O}$ is a high-dimensional space of measurable features

Examples:

- **For a lead:** IP address, device fingerprint, timing, form fields, etc.
- **For a publisher:** Volume, conversion rates, chargeback rates
- **For a consumer:** Browsing behavior, click patterns
- **For a device:** User-agent, screen resolution, JavaScript execution

The Emission Model:

$$O_{i,t} \sim P_{\phi}(\cdot | Z_{i,t})$$

- This is the **likelihood function** in Bayesian inference
- P_{ϕ} is parameterized by ϕ
- Part IV (Chapters 12-15) is dedicated to modeling P_{ϕ}

2.5.3 Fraud Indicators $F(i, t)$

Definition: $F(i, t)$ is the fraud indicator for entity i at time t .

Forms of Fraud Indicators:

- **Binary:** $F(i, t) \in \{0,1\}$ (fraudulent or not)
- **Ordinal:** $F(i, t) \in \{0,1,2,\dots,K\}$ (severity levels)
- **Continuous:** $F(i, t) \in [0,1]$ (fraud probability)

The Fraud Function:

$$F_{i,t} = g(Z_{i,t})$$

- $g: \mathcal{S} \rightarrow \mathcal{F}$ maps hidden state to fraud indicator
- In the simplest case, g is a threshold function
- In general, g can be a complex, learned mapping

Why Separate Z and F ?:

- Z is a rich, multi-dimensional state
- F is a specific function of that state

- This separation allows:
 - The same state to produce different fraud indicators
 - Fraud indicators to be derived from state dynamics
 - The ability to detect fraud before it becomes apparent

2.5.4 State Transitions and Emissions

The Transition Model:

$$Z_{i,t+1} \mid Z_{i,t}, \{Z_{j,t}\}_{j \in \mathcal{N}(i)} \sim P_{\theta}$$

- θ are the transition parameters
- The state at time $t + 1$ depends on:
 - The entity's own state at time t (Markov property)
 - The states of neighboring entities in the graph
- This is a **graph-coupled Markov process**

Coupled Dynamics:

- Quality propagates through the graph
- A publisher's quality affects the leads it generates
- A lead's quality affects the buyer's behavior
- Fraud rings appear as coupled state changes

The Emission Model:

$$O_{i,t} \mid Z_{i,t} \sim P_{\phi}$$

- ϕ are the emission parameters
- Observations are conditionally independent given the state
- This is the standard state-space assumption

The Complete Model:

$$P(Z_{1:T}, O_{1:T}) = P(Z_1) \prod_{t=1}^{T-1} P_{\theta}(Z_{t+1} \mid Z_t) \prod_{t=1}^T P_{\phi}(O_t \mid Z_t)$$

$\underbrace{\hspace{1.5cm}}$ Initial
 $\underbrace{\hspace{1.5cm}}$ Transition
 $\underbrace{\hspace{1.5cm}}$ Emission

This is the **factorization** that enables efficient inference.

2.6 Complete Notation and Symbol Table

This section provides a comprehensive glossary of notation used throughout both volumes.

Symbol	Description	First Used
E	Set of all entities	Chapter 2.3
e_i	The i-th entity	Chapter 2.3

$\tau(e_i)$	Type of entity e_i	Chapter 2.3
G	Graph of the marketplace	Chapter 2.3
R	Set of relations in the graph	Chapter 2.3
$\mathcal{N}(i)$	Neighbors of entity i	Chapter 2.5.4
$Z(i, t)$	Hidden state of entity i at time t	Chapter 2.5.1
$\mathcal{S}$	State space	Chapter 2.5.1
$O(i, t)$	Observation vector for entity i at time t	Chapter 2.5.2
$\mathcal{O}$	Observation space	Chapter 2.5.2
$F(i, t)$	Fraud indicator for entity i at time t	Chapter 2.5.3
$\mathcal{F}$	Fraud indicator space	Chapter 2.5.3
$P_\theta(\cdot \cdot)$	Transition model with parameters θ	Chapter 2.5.4
$P_\phi(\cdot \cdot)$	Emission model with parameters ϕ	Chapter 2.5.4
$g(\cdot)$	Fraud function mapping state to fraud indicator	Chapter 2.5.3
T	Time horizon	Chapter 2.5.4
n	Number of entities	Chapter 2.3
t	Time index	Chapter 2.5.1
θ	Transition model parameters	Chapter 2.5.4
ϕ	Emission model parameters	Chapter 2.5.4
Θ	Parameter space for θ	Chapter 2.5.4
Φ	Parameter space for ϕ	Chapter 2.5.4
$\hat{Z}$	Estimate of hidden state	Chapter 9
α	Regularization parameter	Chapter 12
γ	Discount factor in RL	Chapter 23
π	Policy in RL	Chapter 23
$\mathcal{B}$	Buyer set	Chapter 2.2.6
$\mathcal{P}$	Publisher set	Chapter 2.2.2
$\mathcal{C}$	Consumer set	Chapter 2.2.3
$\mathcal{D}$	Device set	Chapter 2.2.4
$\mathcal{S}_{\text{session}}$	Session set	Chapter 2.2.5
$\mathcal{Cmp}$	Campaign set	Chapter 2.2.7
$\mathcal{A}$	Auction set	Chapter 2.2.8

<i>Conv</i>	Conversion set	Chapter 2.2.9
-------------	----------------	---------------

2.7 The Master Formalism in Words

This section provides an intuitive, non-technical summary of the master formalism.

The Core Idea:

Every participant in the lead marketplace—whether publisher, consumer, device, session, buyer, campaign, or lead itself—has an underlying “quality state” that cannot be directly observed. This state evolves over time and influences everything that can be observed.

The Three Components:

- **Hidden States (Z):**
 - Unobservable quality/trust of each entity
 - Evolves over time
 - Coupled across entities (a publisher’s state affects leads’ states)
- **Observations (O):**
 - Everything we can measure (IP, device fingerprint, conversions, etc.)
 - Generated probabilistically from the hidden state
 - Noisy and imperfect
- **Fraud Indicators (F):**
 - A function of the hidden state
 - What we ultimately want to know

The Inference Problem:

Given observations O , we want to infer the hidden states Z and detect fraud F .

Why This Works:

- State-space models have a rich theoretical foundation
- The separation of state and observation enables principled inference
- The coupled graph structure captures real-world dependencies
- The formalism unifies diverse methods (filtering, ML, game theory)

2.8 Summary and Outlook

Chapter Summary:

- Formalized the lead marketplace as a dynamic probabilistic system
- Defined all entities and their mathematical representations
- Introduced the graph structure of the marketplace
- Established the master state-space model
- Provided a complete notation table

Connections to Subsequent Chapters:

Chapter	Connection
Chapter 3	Probability theory for the state-space model
Chapter 4-8	Fraud and trust as specific applications
Chapter 9-11	Bayesian inference in the state-space model
Chapter 12-15	Modeling P_ϕ and P_θ
Chapter 16-17	Graph and information-theoretic methods
Chapter 18-20	Behavioral models for O
Chapter 21-25	Decision, game, and adversarial theory
Chapter 26-27	Evaluation and causal inference
Chapter 28-29	Synthesis and open problems
Volume II	Systems and AI implementations

2.9 Bibliographic Notes

Bibliographic Notes:

1. Foundational Texts on State-Space Models:

- Rabiner, L. R. (1989). "A tutorial on hidden Markov models and selected applications in speech recognition." *Proceedings of the IEEE*.
- Shumway, R. H., & Stoffer, D. S. (2017). *Time Series Analysis and Its Applications*. Springer.
- Durbin, J., & Koopman, S. J. (2012). *Time Series Analysis by State Space Methods*. Oxford University Press.

2. Graph Theory and Network Science:

- Newman, M. E. J. (2018). *Networks*. Oxford University Press.
- Barabási, A.-L. (2016). *Network Science*. Cambridge University Press.

3. Bayesian Inference:

- Gelman, A., et al. (2013). *Bayesian Data Analysis*. CRC Press.
- Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer.

4. Industry Context:

- Edelman, B., & Schwarz, M. (2017). "Pixels, Prey, and Proxies: The Economics of Ad Fraud." *Harvard Business School Working Paper*.

5. Ecosystem Mapping:

- Lead Gen Ecosystem Overview by B. K. (2023, March 20). *LeadGen Daily*. (This text is referenced as "the original text" in the chapter's source material)

Chapter 3: Probability Theory for Lead Systems

3.1 Random Variables and Sample Spaces

This section establishes the foundational mathematical language for quantifying uncertainty in lead generation systems.

Sample Spaces and Events

Definition: Sample Space (Ω) The sample space is the set of all possible outcomes of a random experiment in the lead ecosystem.

- **Discrete Sample Space:** Finite or countably infinite outcomes
 - Example: $\Omega = \{\text{Lead converts, Lead does not convert}\}$
 - Example: $\Omega = \{0, 1, 2, \dots, 100\}$ (number of leads from a publisher in an hour)
- **Continuous Sample Space:** Uncountably infinite outcomes
 - Example: $\Omega = \mathbb{R}^+$ (time until a lead converts)
 - Example: $\Omega = [0, 1]$ (probability that a lead is fraudulent)

Definition: Event (A) An event is a subset of the sample space, $A \subseteq \Omega$.

- Example: $A = \{\text{Lead converts within 24 hours}\}$
- Example: $B = \{\text{Publisher quality drops below threshold}\}$

Key Insight for Lead Systems: Every transaction, click, form submission, and conversion is a realization of a random experiment. The uncertainty arises from:

- Consumer behavior (unpredictable)
- Adversarial actions (intentionally random or hidden)
- Measurement error (noise in observations)
- Incomplete information (hidden variables)

Random Variables

Definition: Random Variable (X) A function $X: \Omega \rightarrow \mathbb{R}$ that assigns a numerical value to each outcome.

Types of Random Variables in Lead Systems:

Type	Definition	Examples
------	------------	----------

Binary	$X \in \{0,1\}$	Conversion flag, fraud flag
Categorical	$X \in \{1,2,\dots,K\}$	Traffic source (organic, paid, social), buyer vertical
Count	$X \in \mathbb{Z}^+$	Number of leads, number of sessions
Continuous	$X \in \mathbb{R} \text{ or } \mathbb{R}^+$	Lead price, conversion time, quality score

Indicator Random Variables: A special type of binary random variable that indicates whether an event occurred:

$$\mathbb{1}_A = \begin{cases} 1 & \text{if event } A \text{ occurs} \\ 0 & \text{otherwise} \end{cases}$$

Example in Lead Systems: Let $X = \mathbb{1}_{\{\text{Lead converts}\}}$. Then:

- $X = 1$ if the lead converts
- $X = 0$ if the lead does not convert

Distribution Functions

Probability Mass Function (PMF): For discrete X

$$p_X(x) = P(X = x)$$

Probability Density Function (PDF): For continuous X

$$f_X(x) = \frac{d}{dx}P(X \leq x)$$

Cumulative Distribution Function (CDF): For any X

$$F_X(x) = P(X \leq x)$$

Lead System Example: Let X be the number of leads generated by a publisher in an hour. The PMF might be:

- $p_X(0) = 0.1$ (10% chance of zero leads)
- $p_X(1) = 0.3$ (30% chance of one lead)
- $p_X(2) = 0.3$ (30% chance of two leads)
- $p_X(3) = 0.2$ (20% chance of three leads)
- $p_X(x) = 0$ for $x \geq 4$

Measurability and Filtration

Definition: Filtration ($\mathcal{F}_t$) An increasing sequence of σ -algebras representing information available up to time t .

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots \subseteq \mathcal{F}_T$$

In Lead Systems:

- $\mathcal{F}_0$: Information before the lead is generated
- $\mathcal{F}_1$: Information at lead generation (form fields, IP, device)
- $\mathcal{F}_2$: Information after routing through ping tree (bids, price)
- $\mathcal{F}_3$: Information after conversion outcome (profit/loss)

Why This Matters: Fraud detection is a **filtering problem**—we make decisions based on the information available at each time step.

3.2 Conditional Probability and Independence

This section develops the tools for reasoning about relationships between random variables in the lead ecosystem.

Conditional Probability

Definition: Conditional Probability

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

This is the probability of event A given that event B has occurred.

Lead System Example:

- A = “Lead converts to a sale”
- B = “Lead was generated by a high-quality publisher”
- $P(A | B)$ = Conversion probability for leads from high-quality publishers

The Chain Rule: For events $A_1, A_2, \dots, A_n$:

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1) \cdot P(A_2 | A_1) \cdot P(A_3 | A_1 \cap A_2) \cdots P(A_n | A_1 \cap \dots \cap A_{n-1})$$

Application to Lead Lifecycle: Let A_1 = “Consumer clicks ad”, A_2 = “Consumer fills form”, A_3 = “Lead is purchased”, A_4 = “Lead converts”.

$$P(A_1 \cap A_2 \cap A_3 \cap A_4) = P(A_1) \cdot P(A_2 | A_1) \cdot P(A_3 | A_1, A_2) \cdot P(A_4 | A_1, A_2, A_3)$$

Independence

Definition: Independence of Events Events A and B are independent if:

$$P(A \cap B) = P(A) \cdot P(B)$$

Equivalently:

$$P(A | B) = P(A) \quad \text{and} \quad P(B | A) = P(B)$$

Definition: Conditional Independence Events A and B are conditionally independent given C if:

$$P(A \cap B | C) = P(A | C) \cdot P(B | C)$$

or equivalently:

$$P(A | B, C) = P(A | C)$$

Key Insight for State-Space Models: In the master formalism (Chapter 2), observations $O_{1,t}$ and $O_{2,t}$ are conditionally independent given the hidden state Z_t :

$$P(O_{1,t}, O_{2,t} | Z_t) = P(O_{1,t} | Z_t) \cdot P(O_{2,t} | Z_t)$$

This is the **conditional independence assumption** that makes inference tractable.

Conditional Dependence

Example of Conditional Dependence in Lead Systems:

- X = Lead quality (conversion rate)
- Y = Publisher reputation
- Z = Traffic source (organic vs. paid)

X and Y are conditionally independent given Z if publisher reputation only affects conversion through traffic source.

Violation of Conditional Independence: If publisher reputation affects conversion through other mechanisms (e.g., brand trust, landing page quality), then:

$$P(X | Y, Z) \neq P(X | Z)$$

Why This Matters: The conditional independence assumption is the foundation of the master formalism. If violated, the model is misspecified.

3.3 Bayes' Theorem and Its Implications

This section presents the most important theorem for fraud detection and explores its implications.

Bayes' Theorem

Statement: For events A and B with $P(B) > 0$:

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

For Random Variables (Discrete):

$$P(X = x | Y = y) = \frac{P(Y = y | X = x) \cdot P(X = x)}{\sum_{x'} P(Y = y | X = x') \cdot P(X = x')}$$

For Random Variables (Continuous):

$$f_{X|Y}(x | y) = \frac{f_{Y|X}(y | x) \cdot f_X(x)}{\int f_{Y|X}(y | x') \cdot f_X(x') dx'}$$

The Three Components of Bayes' Theorem

1. Prior: $P(A)$ or $P(X = x)$

- The belief about A before observing evidence B
- **In Lead Systems:** The baseline fraud rate, prior belief in publisher quality

2. Likelihood: $P(B | A)$ or $P(Y = y | X = x)$

- The probability of observing the evidence given the state
- **In Lead Systems:** The distribution of observations (IP, device, behavior) given fraud state

3. Posterior: $P(A | B)$ or $P(X = x | Y = y)$

- The updated belief after observing evidence
- **In Lead Systems:** The probability that a lead is fraudulent given all available observations

Applications to Fraud Detection

Example: Basic Fraud Detection Let:

- F = Event that a lead is fraudulent
- E = Event that an observed pattern is suspicious

We want: $P(F | E)$ (posterior probability of fraud given suspicious evidence)

Using Bayes:

$$P(F | E) = \frac{P(E | F) \cdot P(F)}{P(E | F) \cdot P(F) + P(E | \text{not } F) \cdot P(\text{not } F)}$$

Numerical Example:

- $P(F) = 0.02$ (2% baseline fraud rate)
- $P(E | F) = 0.90$ (90% of fraudulent leads show suspicious pattern)
- $P(E | \text{not } F) = 0.05$ (5% of legitimate leads show suspicious pattern)

Then:

$$P(F | E) = \frac{0.90 \cdot 0.02}{0.90 \cdot 0.02 + 0.05 \cdot 0.98} = \frac{0.018}{0.018 + 0.049} = \frac{0.018}{0.067} \approx 0.269$$

Interpretation: Even with suspicious evidence, the lead is only 27% likely to be fraudulent due to the low base rate. This is the **base rate fallacy** in action—and a critical lesson for fraud detection.

Sequential Bayesian Updating

Key Insight: Bayes' theorem can be applied sequentially as new evidence arrives.

Sequential Update Formula:

$$P(Z | O_{1:t}) \propto P(O_t | Z) \cdot P(Z | O_{1:t-1})$$

In Words: The posterior after observing $O_{1:t}$ is proportional to:

- The likelihood of the current observation O_t given Z
- The posterior after observing $O_{1:t-1}$

Why This Matters for Lead Systems:

- Leads generate multiple observations over their lifecycle
- We can update fraud probability as new data arrives
- This is the foundation of real-time fraud detection (Chapter 34)

Bayesian Decision Theory

Optimal Decision Rule: Choose action a that minimizes expected loss:

$$a^* = \underset{a}{\operatorname{argmin}} \mathbb{E}[\text{Loss}(a, Z) \mid O]$$

Example: Fraud Threshold Let:

- a_0 = “Accept lead” (treat as legitimate)
- a_1 = “Reject lead” (treat as fraudulent)

Costs:

- C_{FP} = Cost of false positive (rejecting a good lead)
- C_{FN} = Cost of false negative (accepting a bad lead)

Optimal decision: Accept if

$$P(F \mid O) < \frac{C_{FP}}{C_{FP} + C_{FN}}$$

3.4 Joint Distributions and Factorizations

This section introduces the joint distribution of multiple random variables and the factorization that enables efficient computation.

Joint Distribution

Definition: Joint PMF (Discrete)

$$p_{X,Y}(x, y) = P(X = x, Y = y)$$

Definition: Joint PDF (Continuous)

$$f_{X,Y}(x, y) = \frac{\partial^2}{\partial x \partial y} P(X \leq x, Y \leq y)$$

Definition: Joint CDF

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$$

Marginal Distributions

From Joint to Marginal (Discrete):

$$p_X(x) = \sum_y p_{X,Y}(x, y)$$

From Joint to Marginal (Continuous):

$$f_X(x) = \int f_{X,Y}(x, y) dy$$

Interpretation: Marginalizing “integrates out” the other variables.

Lead System Example: Joint distribution of lead quality (Q) and publisher trust (T):

$$p(Q, T) = \begin{pmatrix} 0.05 & 0.10 & 0.05 \\ 0.10 & 0.20 & 0.10 \\ 0.05 & 0.10 & 0.05 \\ 0.10 & 0.05 & 0.05 \end{pmatrix}$$

Rows: $Q \in \{\text{Good, Fair, Poor, Fraudulent}\}$ Columns: $T \in \{\text{Trusted, Unknown, Suspicious}\}$

Marginal distribution of Q :

$$p_Q(\text{Good}) = 0.05 + 0.10 + 0.05 = 0.20$$

$$p_Q(\text{Fair}) = 0.10 + 0.20 + 0.10 = 0.40$$

$$p_Q(\text{Poor}) = 0.05 + 0.10 + 0.05 = 0.20$$

$$p_Q(\text{Fraudulent}) = 0.10 + 0.05 + 0.05 = 0.20$$

Conditional Distributions from Joint

Definition: Conditional PMF

$$p_{X|Y}(x | y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}$$

Definition: Conditional PDF

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

The Chain Rule of Probability

For Events:

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1) \cdot P(A_2 | A_1) \cdots P(A_n | A_1 \cap \dots \cap A_{n-1})$$

For Random Variables:

$$p(X_1, X_2, \dots, X_n) = p(X_1) \cdot p(X_2 | X_1) \cdots p(X_n | X_1, \dots, X_{n-1})$$

Factorizations

Definition: Factorization A factorization expresses a joint distribution as a product of lower-dimensional conditional distributions.

Example: Naive Bayes Factorization

$$p(X_1, X_2, \dots, X_n | Y) = p(Y) \cdot \prod_{i=1}^n p(X_i | Y)$$

This assumes all features X_i are conditionally independent given Y .

Example: State-Space Model Factorization

$$P(Z_{1:T}, O_{1:T}) = P(Z_1) \cdot \prod_{t=2}^T P(Z_t | Z_{t-1}) \cdot \prod_{t=1}^T P(O_t | Z_t)$$

This is the **factorization** used throughout the book.

Graphical Models

Directed Acyclic Graph (DAG) Representation: A DAG represents a factorization:

- Nodes = Random variables
- Edges = Conditional dependencies
- Factorization = Product of conditional distributions given parents

Bayesian Network Example:

$$p(X_1, X_2, X_3, X_4) = p(X_1) \cdot p(X_2) \cdot p(X_3 | X_1, X_2) \cdot p(X_4 | X_3)$$

Application to Lead Systems: A factor graph for lead generation might include:

- C = Consumer intent (hidden)
- P = Publisher quality (hidden)
- O_{click} = Click observation
- O_{form} = Form submission observation
- $O_{conversion}$ = Conversion observation

Factorization:

$$p(C, P, O_{click}, O_{form}, O_{conversion}) = p(C) \cdot p(P) \cdot p(O_{click} | C) \cdot p(O_{form} | C, P) \cdot p(O_{conversion} | C)$$

3.5 Hidden Variables and Latent States

This section introduces the concept of unobserved variables that must be inferred from observed data.

Definition: Hidden Variable

A **hidden variable** (or latent variable) is a random variable that is not directly observed but influences observed variables.

Formal Definition: Let X be an observed random variable and Z be a hidden random variable. The complete data is (X, Z) , but we only observe X .

Why Hidden Variables Are Necessary

Example 1: Consumer Intent

- We observe clicks, form fills, and browsing behavior
- We want to infer consumer intent (high, medium, low)
- Intent is hidden; behavior is observed

Example 2: Publisher Quality

- We observe lead volume, conversion rates, and chargeback rates
- We want to infer publisher quality (Excellent, Good, Poor, Fraudulent)
- Quality is hidden; performance metrics are observed

Example 3: Fraud State

- We observe IP addresses, device fingerprints, timing patterns
- We want to infer whether a lead is fraudulent
- Fraud state is hidden; observations are noisy

The Hidden Variable Model

General Form:

$$P(X, Z) = P(Z) \cdot P(X | Z)$$

Where:

- $P(Z)$ is the prior distribution over hidden states
- $P(X | Z)$ is the likelihood of observations given hidden state

The Inference Problem:

$$P(Z | X) = \frac{P(X | Z) \cdot P(Z)}{\int P(X | Z') \cdot P(Z') dZ'}$$

This is the **posterior distribution**—the goal of Bayesian inference.

Types of Hidden Variables

Type	Description	Example
------	-------------	---------

Discrete	Hidden variable takes finite/countable values	Publisher quality: Good/Bad
Continuous	Hidden variable takes real values	Quality score: $Z \in [0,1]$
Categorical	Hidden variable takes unordered categories	Traffic source: Organic/Paid/Social
Structural	Hidden variable indicates model structure	Which fraud pattern is active?

Identifiability

Definition: Identifiability A model is identifiable if different parameter values produce different distributions over observations.

Formal Definition: The model is identifiable if:

$$P_{\theta_1}(X) = P_{\theta_2}(X) \Rightarrow \theta_1 = \theta_2$$

Implications for Fraud Detection:

- If the model is not identifiable, we cannot distinguish between different explanations of the data
- Example: Is a low conversion rate due to poor publisher quality or bad campaign targeting?
- Identifiability is essential for causal inference

Example of Non-Identifiability: Let $Z \sim \text{Bernoulli}(p)$ and $X | Z \sim \mathcal{N}(Z\mu, \sigma^2)$. The observed distribution is a mixture of two normals:

$$P(X) = (1 - p)\mathcal{N}(0, \sigma^2) + p\mathcal{N}(\mu, \sigma^2)$$

The parameters (p, μ, σ^2) may not be uniquely identifiable.

Latent State Models

Definition: Latent State A latent state is a hidden variable that evolves over time according to a Markov process.

The State-Space Model (Review from Chapter 2):

$$Z_{t+1} \sim P_{\theta}(\cdot | Z_t) \quad (\text{State Transition})$$

$$X_t \sim P_{\phi}(\cdot | Z_t) \quad (\text{Emission})$$

Why This Matters:

- Latent states allow us to model dynamic processes (publisher quality changing over time)
- The Markov property makes inference tractable
- This is the foundation of Bayesian filtering (Chapter 9)

3.6 Expectation, Variance, and Moments

This section covers the key summaries of probability distributions.

Expectation

Definition: Expected Value

- **Discrete:**

$$\mathbb{E}[X] = \sum_x x \cdot p_X(x)$$

- **Continuous:**

$$\mathbb{E}[X] = \int x \cdot f_X(x) dx$$

Properties of Expectation:

- **Linearity:** $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$
- **Monotonicity:** If $X \leq Y$, then $\mathbb{E}[X] \leq \mathbb{E}[Y]$
- **Law of Total Expectation:** $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X | Y]]$

Lead System Examples:

- Expected revenue per lead: $\mathbb{E}[\text{Revenue}]$
- Expected conversion rate: $\mathbb{E}[\mathbb{1}_{\text{conversion}}] = P(\text{conversion})$
- Expected fraud rate: $\mathbb{E}[F] = P(F = 1)$

Variance

Definition: Variance

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Standard Deviation:

$$\text{SD}(X) = \sqrt{\text{Var}(X)}$$

Properties of Variance:

- $\text{Var}(aX + b) = a^2\text{Var}(X)$
- $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$
- If X and Y are independent, $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$

Law of Total Variance:

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X | Y)] + \text{Var}(\mathbb{E}[X | Y])$$

Lead System Examples:

- Variability in lead price across auctions
- Variability in conversion rates across publishers

- Variability in fraud rates across campaigns

Covariance and Correlation

Definition: Covariance

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

Definition: Correlation

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} \in [-1, 1]$$

Interpretation:

- $\text{Corr}(X, Y) > 0$: Positive relationship (quality and price)
- $\text{Corr}(X, Y) < 0$: Negative relationship (fraud rate and conversion)
- $\text{Corr}(X, Y) = 0$: No linear relationship

Lead System Examples:

- Correlation between publisher quality and lead price
- Correlation between fraud indicators and conversion rates
- Correlation between device fingerprints and fraud probability

Higher-Order Moments

Skewness (Third Moment):

$$\text{Skew}(X) = \mathbb{E} \left[\left(\frac{X - \mu}{\sigma} \right)^3 \right]$$

- > 0 : Right-skewed (long tail of high values)
- < 0 : Left-skewed (long tail of low values)
- **Application:** Lead prices are often right-skewed (few high-value leads)

Kurtosis (Fourth Moment):

$$\text{Kurt}(X) = \mathbb{E} \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right] - 3$$

- > 0 : Heavy tails (more outliers)
- < 0 : Light tails (fewer outliers)
- **Application:** Fraud detection often involves outliers—heavy-tailed distributions

3.7 Information and Entropy

This section introduces information-theoretic concepts for quantifying uncertainty.

Entropy

Definition: Entropy (Discrete)

$$H(X) = - \sum_x p_X(x) \log p_X(x)$$

Units: bits (base 2) or nats (base e)

Interpretation:

- Entropy is the average number of bits needed to encode X
- Higher entropy = more uncertainty
- Maximum entropy = uniform distribution

Lead System Examples:

- Entropy of traffic sources: $H(\text{TrafficSource})$
 - Low entropy = concentrated in few sources (less diversity, potentially fraud risk)
 - High entropy = diverse sources (more resilient)

Definition: Entropy (Continuous - Differential Entropy)

$$h(X) = - \int f_X(x) \log f_X(x) dx$$

Conditional Entropy

Definition: Conditional Entropy

$$H(X | Y) = - \sum_y p_Y(y) \sum_x p_{X|Y}(x | y) \log p_{X|Y}(x | y)$$

Interpretation:

- Average uncertainty in X after observing Y
- $H(X | Y) \leq H(X)$ (observation reduces uncertainty)

Lead System Example:

- X = Fraud status
- Y = Device fingerprint
- $H(X | Y) < H(X)$ if device fingerprints help detect fraud

Mutual Information

Definition: Mutual Information

$$I(X; Y) = \sum_x \sum_y p_{X,Y}(x, y) \log \frac{p_{X,Y}(x, y)}{p_X(x)p_Y(y)}$$

Interpretation:

- Average reduction in uncertainty about X when observing Y
- $I(X; Y) \geq 0$ with equality if X and Y are independent

Relationship to Entropy:

$$I(X; Y) = H(X) - H(X | Y) = H(Y) - H(Y | X)$$

Lead System Examples:

- $I(\text{Fraud}; \text{IP Address})$: How much does IP address reveal about fraud?
- $I(\text{Quality}; \text{Conversion Rate})$: How predictive is conversion rate of quality?
- $I(\text{Publisher}; \text{Buyer})$: How much does publisher identity predict buyer behavior?

Information Gain

Definition: Information Gain

$$IG(X | Y) = H(X) - H(X | Y) = I(X; Y)$$

Application: Feature Selection

- Choose features with highest information gain
- Example: Which device fingerprint features most predict fraud?

KL Divergence

Definition: Kullback-Leibler Divergence

$$D_{KL}(P \parallel Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)}$$

Interpretation:

- Measure of how different Q is from P
- $D_{KL}(P \parallel Q) \geq 0$ with equality if $P = Q$
- Not symmetric: $D_{KL}(P \parallel Q) \neq D_{KL}(Q \parallel P)$

Lead System Example:

- P = Observed traffic distribution
- Q = Expected traffic distribution under legitimate behavior
- $D_{KL}(P \parallel Q)$ = How abnormal is the traffic pattern?

Application to Fraud:

$$D_{KL}(\text{Fraudulent distribution} \parallel \text{Legitimate distribution})$$

Measures how distinguishable fraud is from legitimate behavior.

Jensen-Shannon Divergence

Definition: Jensen-Shannon Divergence

$$D_{JS}(P \parallel Q) = \frac{1}{2}D_{KL}(P \parallel M) + \frac{1}{2}D_{KL}(Q \parallel M)$$

where $M = \frac{1}{2}(P + Q)$.

Properties:

- Symmetric: $D_{JS}(P \parallel Q) = D_{JS}(Q \parallel P)$
- Bounded: $0 \leq D_{JS}(P \parallel Q) \leq \log 2$

Application:

- Measuring divergence between publisher traffic distributions
- Detecting changes in behavior over time

3.8 Common Distributions in Fraud Modeling

This section presents the probability distributions most commonly used in fraud detection.

Bernoulli Distribution

PDF/PMF:

$$P(X = x) = p^x(1 - p)^{1-x}, \quad x \in \{0,1\}$$

Parameters:

- $p \in [0,1]$: Probability of success

Moments:

- $\mathbb{E}[X] = p$
- $\text{Var}(X) = p(1 - p)$

Applications:

- Fraud indicator: $F \sim \text{Bernoulli}(p_{\text{fraud}})$
- Conversion indicator: $C \sim \text{Bernoulli}(p_{\text{conv}})$

Binomial Distribution

PDF/PMF:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n$$

Parameters:

- $n \in \mathbb{Z}^+$: Number of trials
- $p \in [0,1]$: Success probability

Moments:

- $\mathbb{E}[X] = np$
- $\text{Var}(X) = np(1 - p)$

Applications:

- Number of fraudulent leads in a batch of n leads
- Number of conversions from n leads

Poisson Distribution

PDF/PMF:

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

Parameters:

- $\lambda > 0$: Average rate

Moments:

- $\mathbb{E}[X] = \lambda$
- $\text{Var}(X) = \lambda$

Applications:

- Number of leads generated in a time period
- Number of clicks on an ad campaign

Gaussian (Normal) Distribution

PDF:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}$$

Parameters:

- $\mu \in \mathbb{R}$: Mean
- $\sigma^2 > 0$: Variance

Moments:

- $\mathbb{E}[X] = \mu$
- $\text{Var}(X) = \sigma^2$

Applications:

- Log-transformed lead prices (approximately normal)
- Measurement errors and noise
- Latent continuous quality scores

Logistic Distribution

PDF:

$$f_X(x) = \frac{e^{-(x-\mu)/s}}{s(1 + e^{-(x-\mu)/s})^2}, \quad x \in \mathbb{R}$$

Parameters:

- $\mu \in \mathbb{R}$: Location
- $s > 0$: Scale

Applications:

- Latent variables for logistic regression
- Probability transformations: $P(\text{fraud}) = \frac{1}{1+e^{-x}}$

Exponential Distribution

PDF:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

Parameters:

- $\lambda > 0$: Rate parameter

Moments:

- $\mathbb{E}[X] = 1/\lambda$
- $\text{Var}(X) = 1/\lambda^2$

Applications:

- Time between events (leads, conversions)
- Time until a publisher is flagged for fraud

Gamma Distribution

PDF:

$$f_X(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x \geq 0$$

Parameters:

- $\alpha > 0$: Shape parameter
- $\beta > 0$: Rate parameter

Moments:

- $\mathbb{E}[X] = \alpha/\beta$
- $\text{Var}(X) = \alpha/\beta^2$

Applications:

- Prior distribution for Poisson rate λ
- Modeling lead prices (right-skewed)
- Modeling wait times

Beta Distribution

PDF:

$$f_X(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad x \in [0,1]$$

Parameters:

- $\alpha > 0, \beta > 0$: Shape parameters

Moments:

- $\mathbb{E}[X] = \frac{\alpha}{\alpha + \beta}$
- $\text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$

Applications:

- Prior for probability p (Bernoulli, Binomial)
- Prior for conversion rates
- Prior for fraud rates

Mixture Models

Definition: Mixture Distribution

$$f_X(x) = \sum_{k=1}^K \pi_k f_k(x; \theta_k)$$

where:

- $\pi_k > 0, \sum_k \pi_k = 1$ are mixing weights
- f_k are component distributions
- θ_k are component parameters

Application: Fraud Mixture Model Let:

- Component 1 = Legitimate leads: $f_1(x; \theta_1)$
- Component 2 = Fraudulent leads: $f_2(x; \theta_2)$
- $\pi_1 = 1 - p_{\text{fraud}}, \pi_2 = p_{\text{fraud}}$

Then:

$$f_X(x) = (1 - p_{\text{fraud}})f_{\text{legit}}(x) + p_{\text{fraud}}f_{\text{fraud}}(x)$$

This is a **mixture model** for fraud detection.

3.9 Worked Applications in Lead Marketplaces

This section applies the probability theory to concrete lead system scenarios.

Application 1: Estimating Fraud Rate

Problem: A lead buyer wants to estimate the fraud rate for a publisher. The buyer observes 1,000 leads and finds that 50 are labeled as fraudulent (via manual review).

Prior: The buyer has a prior belief that the fraud rate is around 3% with uncertainty:

$$p \sim \text{Beta}(\alpha = 3, \beta = 97)$$

(Mean = 3%, effective sample size = 100)

Likelihood: Observing $x = 50$ frauds out of $n = 1000$:

$$P(X = 50 | p) = \binom{1000}{50} p^{50} (1 - p)^{950}$$

Posterior:

$$p | X \sim \text{Beta}(\alpha + x, \beta + n - x) = \text{Beta}(53, 947)$$

Posterior Mean:

$$\mathbb{E}[p | X] = \frac{53}{53 + 947} = \frac{53}{1000} = 0.053$$

Credible Interval (95%):

$$\text{Beta}(53, 947) \text{ 95\% interval } \approx [0.039, 0.068]$$

Interpretation: The estimated fraud rate is 5.3% with 95% confidence interval [3.9%, 6.8%]. The buyer should be concerned—this exceeds the 3% baseline.

Application 2: Sequential Fraud Detection

Problem: A fraud detection system processes leads sequentially. Each lead generates a score between 0 and 1 indicating fraud probability. The system must decide when to stop and flag a lead as fraudulent.

Model:

- Z_t = Hidden state at time t (Legitimate or Fraudulent)
- O_t = Observation at time t (score between 0 and 1)

Emission Model:

- If legitimate: $O_t \sim \text{Beta}(\alpha = 2, \beta = 8)$ (scores concentrated near 0)
- If fraudulent: $O_t \sim \text{Beta}(\alpha = 8, \beta = 2)$ (scores concentrated near 1)

E.4 Information-Theoretic Inequalities

Inequality	Formula
Data Processing	$I(X; Y) \geq I(X; Z)$ if $X \rightarrow Y \rightarrow Z$
Chain Rule	$I(X_1, \dots, X_n; Y) = \sum_i I(X_i; Y \mid X_{i-1}, \dots, X_1)$
Mutual Information Bound	$I(X; Y) \leq H(X)$
KL Divergence	$D_{KL}(P \parallel Q) \geq 0$

Appendix F: Optimization Methods

F.1 Convex Optimization

Definitions

Definition F.1 (Convex Function): A function f is convex if:

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y), \quad \theta \in [0,1]$$

Definition F.2 (Convex Optimization Problem):

$$\min_{x \in \mathcal{C}} f(x)$$

where f is convex and $\mathcal{C}$ is convex.

Properties

Property	Description
Local Optimum	Any local optimum is global
First-Order Condition	$\nabla f(x^*) = 0$
Second-Order Condition	$\nabla^2 f(x^*) \succcurlyeq 0$

F.2 Gradient Descent

Basic Algorithm

Algorithm F.1 (Gradient Descent):

Initialize x_0 For $k = 0, 1, 2, \dots$:

$$x_{k+1} = x_k - \eta_k \nabla f(x_k)$$

Convergence

Condition:

$$\lim_{k \rightarrow \infty} \|\nabla f(x_k)\| = 0$$

Rate:

$$f(x_k) - f(x^*) \leq O(1/k)$$

F.3 Stochastic Optimization

Stochastic Gradient Descent

Algorithm F.2 (SGD):

Initialize x_0 For $k = 0, 1, 2, \dots$:

$$x_{k+1} = x_k - \eta_k \nabla \hat{f}(x_k)$$

where $\nabla \hat{f}$ is an unbiased estimate of ∇f .

Adam Optimizer

Algorithm F.3 (Adam):

$$m_k = \beta_1 m_{k-1} + (1 - \beta_1) \nabla \hat{f}(x_k)$$

$$v_k = \beta_2 v_{k-1} + (1 - \beta_2) (\nabla \hat{f}(x_k))^2$$

$$\hat{m}_k = \frac{m_k}{1 - \beta_1^k}, \quad \hat{v}_k = \frac{v_k}{1 - \beta_2^k}$$

$$x_{k+1} = x_k - \eta \frac{\hat{m}_k}{\sqrt{\hat{v}_k} + \epsilon}$$

F.4 Constrained Optimization

Lagrange Multipliers

Definition F.3 (Lagrangian): For $\min f(x)$ subject to $g_i(x) = 0$:

$$\mathcal{L}(x, \lambda) = f(x) + \sum_i \lambda_i g_i(x)$$

KKT Conditions:

$$\nabla_x \mathcal{L}(x^*, \lambda^*) = 0$$

$$g_i(x^*) = 0$$

F.5 Bayesian Optimization

Gaussian Process Regression

Definition F.4 (GP): A Gaussian process is a distribution over functions:

$$f(x) \sim \mathcal{GP}(m(x), k(x, x'))$$

Acquisition Functions

Function	Formula	Use Case
Expected Improvement	$EI(x) = \mathbb{E}[\max(0, f(x) - f^+)]$	Exploration
Upper Confidence Bound	$UCB(x) = \mu(x) + \kappa\sigma(x)$	Balanced
Probability of Improvement	$PI(x) = P(f(x) > f^+)$	Conservative

Appendix G: Python Primer

G.1 Basic Python

Data Types

Type	Example	Description
int	42	Integer
float	3.14	Floating point
str	"hello"	String
bool	True	Boolean
list	[1, 2, 3]	Mutable sequence
tuple	(1, 2, 3)	Immutable sequence
dict	{"key": "value"}	Key-value pairs
set	{1, 2, 3}	Unique elements

Control Flow

```
# if-else
if condition:
    # do something
elif condition:
    # do something else
else:
    # default
```

```
# for loop
for item in iterable:
    # process item
```

```
# while loop
while condition:
    # do something
```

```
# list comprehension
squares = [x**2 for x in range(10)]
```

Functions

```
def function_name(param1, param2, default=10):  
    """Docstring: Describe function."""  
    result = param1 + param2 + default  
    return result
```

Lambda functions

```
add = lambda x, y: x + y
```

G.2 NumPy

Arrays

```
import numpy as np
```

Create arrays

```
a = np.array([1, 2, 3])           # 1D array  
b = np.array([[1, 2], [3, 4]])   # 2D array  
zeros = np.zeros((3, 4))         # Array of zeros  
ones = np.ones((2, 3))           # Array of ones  
eye = np.eye(5)                  # Identity matrix
```

Array operations

```
c = a + b                         # Element-wise addition  
d = np.dot(a, b)                  # Matrix multiplication  
e = a * b                         # Element-wise multiplication
```

Statistics

Basic statistics

```
mean = np.mean(a)  
std = np.std(a)  
var = np.var(a)  
min_val = np.min(a)  
max_val = np.max(a)
```

Random numbers

```
uniform = np.random.uniform(0, 1, 100)  
normal = np.random.normal(0, 1, 100)
```

G.3 Pandas

DataFrames

```
import pandas as pd
```

Create DataFrame

```
df = pd.DataFrame({  
    'col1': [1, 2, 3],  
    'col2': ['a', 'b', 'c'],  
    'col3': [True, False, True]})
```

```

}))

# Read data
df = pd.read_csv('data.csv')
df = pd.read_parquet('data.parquet')

Data Manipulation
# Filtering
filtered = df[df['col1'] > 1]

# Grouping
grouped = df.groupby('col2').mean()

# Aggregation
agg = df.agg({
    'col1': ['mean', 'std'],
    'col2': 'count'
})

# Merge
merged = pd.merge(df1, df2, on='key')

# Time series
df['date'] = pd.to_datetime(df['date'])
df.set_index('date', inplace=True)
df.resample('D').mean()

```

G.4 Scikit-Learn

Model Training

```

from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split

```

Split data

```

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

```

Scale data

```

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

```

Train model

```

model = RandomForestClassifier(n_estimators=100)
model.fit(X_train_scaled, y_train)

```

```

# Predict
y_pred = model.predict(X_test_scaled)
y_proba = model.predict_proba(X_test_scaled)

Metrics
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score,
    f1_score, roc_auc_score, confusion_matrix
)

# Calculate metrics
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
auc = roc_auc_score(y_test, y_proba[:, 1])
conf_matrix = confusion_matrix(y_test, y_pred)

```

G.5 PyTorch

Tensors

```
import torch
```

Create tensors

```

x = torch.tensor([1, 2, 3])
y = torch.zeros(3, 4)
z = torch.ones(2, 3)

```

Tensor operations

```

w = x + y
v = torch.matmul(x, y.T)

```

Neural Networks

```
import torch.nn as nn
```

Define model

```

class FraudModel(nn.Module):
    def __init__(self, input_dim, hidden_dim, output_dim):
        super().__init__()
        self.fc1 = nn.Linear(input_dim, hidden_dim)
        self.fc2 = nn.Linear(hidden_dim, output_dim)
        self.relu = nn.ReLU()
        self.sigmoid = nn.Sigmoid()

    def forward(self, x):
        x = self.relu(self.fc1(x))
        x = self.sigmoid(self.fc2(x))
        return x

```

Training Loop

```
model = FraudModel(input_dim=10, hidden_dim=64, output_dim=1)
optimizer = torch.optim.Adam(model.parameters())
loss_fn = nn.BCELoss()

for epoch in range(100):
    optimizer.zero_grad()
    pred = model(X)
    loss = loss_fn(pred, y)
    loss.backward()
    optimizer.step()
```

G.6 TensorFlow

Tensors

```
import tensorflow as tf
```

Create tensors

```
x = tf.constant([1, 2, 3])
y = tf.zeros((3, 4))
z = tf.ones((2, 3))
```

Tensor operations

```
w = x + y
v = tf.matmul(x, tf.transpose(y))
```

Keras Models

```
import tensorflow as tf
from tensorflow import keras
```

Sequential model

```
model = keras.Sequential([
    keras.layers.Dense(64, activation='relu', input_shape=(10,)),
    keras.layers.Dropout(0.2),
    keras.layers.Dense(32, activation='relu'),
    keras.layers.Dense(1, activation='sigmoid')
])
```

Compile

```
model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)
```

Train

```
history = model.fit(
    X_train, y_train,
    epochs=100,
```

```
    batch_size=32,  
    validation_data=(X_test, y_test)  
)
```

Appendix H: SQL Examples

H.1 Basic Queries

SELECT Statements

-- Select all columns

```
SELECT * FROM fraud_events;
```

-- Select specific columns

```
SELECT event_id, event_time, ip, is_fraud FROM fraud_events;
```

-- Select with conditions

```
SELECT * FROM fraud_events  
WHERE is_fraud = 1  
AND event_time > '2024-01-01';
```

-- Select with ordering

```
SELECT * FROM fraud_events  
ORDER BY event_time DESC  
LIMIT 100;
```

H.2 Aggregate Functions

-- Count rows

```
SELECT COUNT(*) FROM fraud_events;
```

-- Sum of values

```
SELECT SUM(revenue) FROM fraud_events;
```

-- Average

```
SELECT AVG(quality_score) FROM fraud_events;
```

-- Group by

```
SELECT  
    publisher_id,  
    COUNT(*) as lead_count,  
    AVG(quality_score) as avg_quality  
FROM fraud_events  
GROUP BY publisher_id;
```

-- Having clause

```
SELECT  
    publisher_id,  
    COUNT(*) as lead_count
```

```

FROM fraud_events
GROUP BY publisher_id
HAVING COUNT(*) > 1000;

```

H.3 Window Functions

-- Row number

```

SELECT
    event_id,
    publisher_id,
    ROW_NUMBER() OVER (PARTITION BY publisher_id ORDER BY event_time) as row_num
FROM fraud_events;

```

-- Ranking

```

SELECT
    publisher_id,
    revenue,
    RANK() OVER (ORDER BY revenue DESC) as revenue_rank
FROM publishers;

```

-- Moving average

```

SELECT
    event_time,
    quality_score,
    AVG(quality_score) OVER (
        ORDER BY event_time
        ROWS BETWEEN 7 PRECEDING AND CURRENT ROW
    ) as moving_avg
FROM fraud_events;

```

H.4 Joins

-- Inner join

```

SELECT
    l.event_id,
    p.publisher_name,
    l.quality_score
FROM leads l
INNER JOIN publishers p ON l.publisher_id = p.publisher_id;

```

-- Left join

```

SELECT
    l.*,
    c.conversion_status
FROM leads l
LEFT JOIN conversions c ON l.lead_id = c.lead_id;

```

-- Multiple joins

```

SELECT
    l.*,

```

```

        p.publisher_name,
        b.buyer_name
FROM leads l
JOIN publishers p ON l.publisher_id = p.publisher_id
JOIN buyers b ON l.buyer_id = b.buyer_id;

```

H.5 Subqueries

-- Subquery in WHERE

```

SELECT * FROM fraud_events
WHERE publisher_id IN (
    SELECT publisher_id FROM publishers
    WHERE quality_score < 0.5
);

```

-- Subquery in FROM

```

SELECT
    publisher_id,
    AVG(quality_score) as avg_quality
FROM (
    SELECT * FROM fraud_events
    WHERE event_time > '2024-01-01'
) recent_events
GROUP BY publisher_id;

```

-- Correlated subquery

```

SELECT
    e1.*,
    (SELECT AVG(quality_score)
     FROM fraud_events e2
     WHERE e2.publisher_id = e1.publisher_id
    ) as publisher_avg
FROM fraud_events e1;

```

Appendix I: Statistical Tables

I.1 Normal Distribution

Standard Normal Distribution $\Phi(z) = P(Z \leq z)$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224

Critical Values

Confidence Level	α	$z_{\alpha/2}$
90%	0.10	1.645
95%	0.05	1.960
99%	0.01	2.576
99.9%	0.001	3.291

I.2 t-Distribution

Critical Values $t_{\alpha, \nu}$

ν	0.10	0.05	0.025	0.01	0.005
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
10	1.372	1.812	2.228	2.764	3.169
20	1.325	1.725	2.086	2.528	2.845
30	1.310	1.697	2.042	2.457	2.750
50	1.299	1.676	2.009	2.403	2.678
∞	1.282	1.645	1.960	2.326	2.576

I.3 Chi-Square Distribution

Critical Values $\chi^2_{\alpha, \nu}$

ν	0.95	0.90	0.10	0.05	0.01
1	0.004	0.016	2.706	3.841	6.635
2	0.103	0.211	4.605	5.991	9.210
3	0.352	0.584	6.251	7.815	11.345
4	0.711	1.064	7.779	9.488	13.277
5	1.145	1.610	9.236	11.070	15.086
10	3.940	4.865	15.987	18.307	23.209
20	10.851	12.443	28.412	31.410	37.566

I.4 F-Distribution

Critical Values F_{α, ν_1, ν_2}

ν_1	ν_2	0.95	0.90	0.05	0.01
---------	---------	------	------	------	------

1	1	0.000	0.000	161.45	4052.18
1	10	0.004	0.004	4.965	10.044
1	20	0.004	0.004	4.351	8.096
2	10	0.049	0.049	4.103	7.559
2	20	0.048	0.048	3.493	5.849
5	10	0.186	0.186	3.326	5.636
5	20	0.174	0.174	2.711	4.103

Appendix J: Dataset Descriptions

J.1 Public Fraud Datasets

Dataset	Description	Size	Features	Use Case
Credit Card Fraud	Credit card transactions	284,807	31	Binary classification
IEEE-CIS	Fraud detection	590,540	400+	Binary classification
AMLSim	Money laundering	10,000+	Variable	AML detection
Real-world Online Payment	Online payments	6,000,000	10	Binary classification

J.2 Lead Generation Datasets

Dataset	Description	Size	Features	Use Case
Lead Conversion	Lead quality	10,000+	50+	Quality prediction
Publisher Quality	Publisher metrics	1,000+	30+	Quality assessment
Ping Tree Data	Auction data	100,000+	20+	Pricing analysis

J.3 Synthetic Data Generation

```
def generate_fraud_dataset(n_samples=10000, fraud_rate=0.05):
    """
    Generate synthetic fraud detection dataset.
    """
    # Generate legitimate data
    n_legit = int(n_samples * (1 - fraud_rate))
    legit = {
        'ip_reputation': np.random.normal(0.8, 0.1, n_legit),
        'device_trust': np.random.normal(0.7, 0.15, n_legit),
        'session_duration': np.random.exponential(60, n_legit),
        'is_fraud': np.zeros(n_legit)
    }

    # Generate fraudulent data
```

```

n_fraud = n_samples - n_legit
fraud = {
    'ip_reputation': np.random.normal(0.2, 0.15, n_fraud),
    'device_trust': np.random.normal(0.3, 0.2, n_fraud),
    'session_duration': np.random.exponential(10, n_fraud),
    'is_fraud': np.ones(n_fraud)
}

# Combine
data = {k: np.concatenate([legit[k], fraud[k]])
        for k in legit.keys()}

return pd.DataFrame(data)

```

J.4 Data Format and Schema

Lead Event Schema

```

{
    "event_id": "evt_123456",
    "event_type": "lead_generation",
    "timestamp": "2024-01-15T10:23:45Z",
    "lead_id": "lead_789",
    "publisher_id": "pub_456",
    "buyer_id": "buyer_123",
    "ip": "192.168.1.1",
    "device_fingerprint": "fp_abc123",
    "session_id": "sess_456",
    "email": "user@domain.com",
    "phone": "+15551234567",
    "quality_score": 0.85,
    "is_fraud": 0,
    "revenue": 10.00
}

```

Appendix K: Proofs of Key Theorems

K.1 Bayesian Optimality

Theorem K.1 (Bayesian Optimality): The Bayes classifier minimizes expected loss.

Proof:

Let $\delta(x)$ be a decision rule. The expected loss is:

$$R(\delta) = \mathbb{E}_{X,Z}[\ell(Z, \delta(X))]$$

By iterated expectation:

$$R(\delta) = \mathbb{E}_X[\mathbb{E}_{Z|X}[\ell(Z, \delta(X))]]$$

For each x , the optimal decision minimizes:

$$\delta^*(x) = \operatorname{argmin}_a \mathbb{E}_{Z|X=x}[\ell(Z, a)]$$

Thus, δ^* minimizes $R(\delta)$.

K.2 Consistency of Estimators

Theorem K.2 (MLE Consistency): Under regularity conditions, the MLE is consistent.

Proof:

Let $\hat{\theta}_n = \operatorname{argmax}_{\theta} \mathcal{L}_n(\theta)$.

The log-likelihood is:

$$\ell_n(\theta) = \frac{1}{n} \sum_{i=1}^n \log f(X_i | \theta)$$

By the Law of Large Numbers:

$$\ell_n(\theta) \rightarrow \mathbb{E}[\log f(X | \theta)] = \ell(\theta) \quad (\text{in probability})$$

By information inequality, $\ell(\theta)$ is uniquely maximized at θ_0 . Thus:

$$\hat{\theta}_n \rightarrow \theta_0 \quad (\text{in probability})$$

K.3 Identifiability Results

Theorem K.3 (Identifiability of Hidden Markov Models): Under certain conditions, HMM parameters are identifiable up to permutation.

Proof Sketch:

1. The transition matrix A and emission matrix B determine the joint distribution
2. The joint distribution determines the spectral decomposition
3. The spectral decomposition determines A and B up to permutation
4. With constraints (e.g., ordering), parameters are unique

The key result is:

$$P(O_{1:T}) = \pi B A B A \cdots B \mathbf{1}$$

By matching moments, parameters are identified.

K.4 Convergence Rates

Theorem K.4 (Parametric Convergence Rate): Under regularity conditions, MLE achieves $1/\sqrt{n}$ convergence rate.

Proof:

The MLE expansion gives:

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = -\frac{1}{\sqrt{n}} \sum_{i=1}^n \ell'_{\theta}(X_i, \theta_0) \cdot I(\theta_0)^{-1} + o_p(1)$$

By CLT:

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \ell'_{\theta}(X_i, \theta_0) \rightarrow \mathcal{N}(0, I(\theta_0)) \quad (\text{in distribution})$$

Thus:

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \rightarrow \mathcal{N}(0, I(\theta_0)^{-1}) \quad (\text{in distribution})$$

Therefore:

$$\|\hat{\theta}_n - \theta_0\| = O_p(n^{-1/2})$$

K.5 Asymptotic Normality

Theorem K.5 (Asymptotic Normality of MLE):

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \rightarrow \mathcal{N}(0, I(\theta_0)^{-1}) \quad (\text{in distribution})$$

Proof:

From the MLE expansion:

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \ell'_{\theta}(X_i, \theta_0) \cdot I(\theta_0)^{-1} + o_p(1)$$

By the CLT:

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \ell'_{\theta}(X_i, \theta_0) \rightarrow \mathcal{N}(0, I(\theta_0)) \quad (\text{in distribution})$$

Therefore:

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \rightarrow \mathcal{N}(0, I(\theta_0)^{-1}) \quad (\text{in distribution})$$